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Journal of Econometrics

journal homepage: www.elsevier.com/locate/jeconom

Identification-robust and simultaneous inference in multifactor asset pricing models[☆]

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ARTICLE INFO

JEL classification:

C3
C58
G11
G12

MSC:

91
62F40
62H15
62P05
91G10
91G15

Keywords:

Linear asset pricing
Traded and non-traded factors
Weak identification
Identification-robust inference
Multivariate linear regression
Exact test

ABSTRACT

This paper proposes exact identification-robust confidence sets for the zero-beta rate and ex-post factor prices in asset pricing models. Exploiting the information from the cross-sectional intercept allows us to impose or formally test model-consistent restrictions, including those resulting from traded factors in excess of the zero beta-rate or from return spreads. Analytical projection-based solutions for confidence set outcomes are developed. The proposed procedures are extended to the case of missing factors. Empirical and simulation results with traded and non-traded factors show that model-consistent restrictions and elusive factors can materially affect model fit, identification, inference and temporal constancy of pricing influence.

[☆] This work was supported by the Social Sciences and Humanities Research Council of Canada, Canada, the Natural Sciences and Engineering Research Council of Canada, the Fonds FQRSC and NATECH (Government of Québec), the Chaire RBC en innovations financières (Université Laval) and the William Dow Chair of Political Economy (McGill University).

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1. Introduction

Linear multi-factor asset pricing models (LAPM) are core components of empirical analysis, whereby expected returns are assumed to be characterized by multiple factors.¹ Despite the wide variety of available candidate factors, the driving premise since Fama and French (1993) [reiterated in Fama and French (2018)] is that the number of relevant factors must be limited. These factors are often themselves returns on a *tradable* portfolio, like the market portfolio, or reflect *non-tradable* forces, like consumption, inflation or liquidity. While most of this literature has focused on factor choice, estimating and testing a given specification still raises challenges. The main difficulty is to assess whether proposed variables are indeed priced factors. This paper proposes finite-sample identification-robust inference methods for such purposes, which discriminate between tradable and non-tradable factors and allow for missing factors.

LAPMs involve nonlinear relations which characterize factor prices *jointly*, along with the zero-beta rate, factor expectations and unknown factor loadings (the so-called factor *betas*); see Campbell et al. (1997, Chapter 6). The literature now recognizes associated identification problems. Traditionally, these relations have been estimated using two-pass methods [reviewed by Shanken and Zhou (2007)], where the first pass uses time series regressions of returns on factors, to estimate the *betas*, and the second pass involves cross-sectional regressions of returns on the estimated *betas*, to obtain factor prices. Consequently, measurement errors arising from estimated *betas* have long been considered a central identification issue. A recent research strand highlights deeper problems resulting from insignificant or homogeneous *betas*; see Kan and Zhang (1999), Beaulieu et al. (2009, 2013), Kleibergen (2009), Kan et al. (2013), Gospodinov et al. (2014), Kleibergen and Zhan (2015), Beaulieu et al. (2016), Gospodinov et al. (2017), Kleibergen et al. (2021) and Kleibergen and Zhan (2020). Generally, the above work shows that factor prices are not identified unless the true (unknown) matrix of *betas* has full rank.

Identification problems thus affect multiple parameters and may have several sources. The underlying nonlinear relations depend on whether the considered factors are tradable or not. Despite well-known advantages, focus on traded factors is unduly restrictive; see Shanken and Weinstein (2006), Shanken and Zhou (2007), and the above cited literature on competing factors. For a discussion of the advantages of traded factors, see Gospodinov et al. (2019), Barillas and Shanken (2017, 2018), and Pukthuanthong et al. (2019). Concretely, implications of traded factors have been operationalized by testable restrictions involving the zero-beta rate (Barone-Adesi et al., 2004; Penaranda and Sentana, 2016). However, empirical analysts often sidestep cross-sectional intercepts (Lewellen et al., 2010), thereby forfeiting important pricing relations.

In view of the above considerations, this paper makes three contributions. *First*, we propose exact identification-robust confidence sets for the zero-beta rate and factor prices, which preserve the association between all LAPM parameters, including realized or sample dependent ones. These sets allow one to assess whether candidate factors command a premium. We focus on a parameter vector introduced by Shanken (1985, 1992) as the ex-post price, which has recently regained interest. This parameter uses the discrepancy between the empirical factor mean and its expectation, which is relevant from a finite-sample perspective; see Khalaf and Schaller (2016), Jegadeesh et al. (2019) and Kim and Skoulakis (2018).

Methodologically, we extend the identification-robust sets proposed in Beaulieu et al. (2013) – for a model with only one tradable factor – in order to allow for models with multiple factors that may be non-tradable. In this case, the number of parameters to identify increases with the number of non-traded factors. Crucially, we exploit the information content of the cross-sectional intercept to uncover links that are lost when returns are transformed to deviations from some asset [as in Kleibergen (2009) and Kleibergen and Zhan (2015), Kleibergen et al. (2021) and Kleibergen and Zhan (2020)]. To fill this gap, we use a simultaneous identification-robust approach which provides information on factor price jointly with the zero-beta rate. We can thus extract information on the latter parameter, which provides assessments of tradable (vs. non-tradable) factor restrictions. This approach is new to the literature, and is distinct through our focus on ex-post parameters.

From a numerical viewpoint, we give analytical solutions which allow one to easily compute the proposed confidence sets. As appropriate when identification may be weak, these are built through test inversions. In contrast with Kan et al. (2013) and Gospodinov et al. (2014), the underlying statistics are not *t*-type measures. Instead, we use Hotelling-type statistics [see Hotelling (1947), Dufour and Khalaf (2002)]. We show that resulting inversions lead to multidimensional quadratic inequalities, and provide tractable analytical solutions building on the mathematics of quadrics [used in Dufour and Taamouti (2005) within a substantially different framework]. The resulting sets can be unbounded or empty, reflecting identification failure or misspecification.

The current state of the literature strongly suggests the need to detect and address misspecification. Given the proliferation of candidate factors, it is unlikely that any given specification will be immune to missing factors. In this context, the *second* contribution of this paper is an extension of our procedures to the case of missing factors. To do this, we adopt the framework from Stambaugh and Yuan (2017) which can be traced back to MacKinlay and Pastor (2000) and MacKinlay (1995), whereby mispricing is embedded within a complete LAPM model albeit with an unobserved orthogonal factor. This approach is distinct because it preserves LAPM parameters and pricing structure. Instead, Almeida and Garcia (2012), Kan et al. (2013) and Gospodinov et al. (2014) introduce pseudo-true price parameters which minimize a distance measure between pricing errors and a zero-expectation. Pseudo-true values conveniently pave the way towards conventional econometric methods, though at the expense of relaxing the LAPM structure. While reasonable departures from the latter structure provide useful misspecification concepts, we adopt the missing factor alternative as

¹ The literature is far too broad to be reviewed exhaustively; see for example Lewellen et al. (2010) on the need to raise the bar in empirical assessments, Harvey et al. (2016) and an update by Harvey and Liu (2019), Barillas and Shanken (2018), Hou et al. (2018), Ahmed et al. (2019), and the popular study by Feng et al. (2020), which has led to describing the current state of the literature as a “factor zoo”.

defined by MacKinlay (1995) and frame the tests underlying our proposed confidence set conformably. With no further data or a priori restrictions, we show that recasting our proposed Hotelling tests in this form allows one to derive finite-sample bounds on their null distribution, which lead to valid (conservative) confidence sets. These sets, again, are characterized by quadratic inequalities with numerically simple analytical projections. The novelty of our approach lies in the fact that we preserve the original ex-post price and zero-beta rate. Furthermore, since the error component we introduce is not restricted beyond its ability to restore exact LAPM pricing, our parsimonious approach can accommodate a wide range of anomalies or other mispricing effects.

The *third* contribution of our paper is an empirical analysis, which we supplement with an illustrative simulation study. We first analyse a model which includes the market factor, a non-tradable liquidity factor (Pástor and Stambaugh, 2003) and an oil price factor (Chen et al., 1986; Ferson and Harvey, 1993; Shanken and Weinstein, 2006), with and without momentum (Carhart, 1997). We next consider the Fama–French three-factor model (Fama and French, 1992, 2018), with and without momentum or non-traded factors. Our analysis accounts for model-consistent restrictions which stem from the nature of the factors; these include the fact that some are traded or are returns spreads (Shanken and Weinstein, 2006).

Key findings can be summarized as follows. (1) Simulation results show the power advantages of our approach which gleans information from the cross-sectional intercept on model-consistent restrictions. (2) Taking these restrictions into account can materially affect model fit, identification, inference and temporal stability of pricing influence. As a consequence, confidence sets reveal concrete trade-offs between identification and model fit. (3) When treated as a complete specification with unrestricted size and book-to-market premiums, results on the Fama–French model underscore the relevance of the market factor. (4) The oil price, liquidity and momentum factors have time varying identification strength. The essence of the problem, particularly for oil and liquidity, is the scarcity of extreme events which drive identification. That result enhances the evidence that factors can be dormant for lengthy periods but awake upon important shocks. It is thus difficult to identify their effects. (5) While elusive factors² may thus lead us to expect an unstable risk return relation, the Fama–French model augmented with liquidity emerges as the best specification from a long-run perspective. (6) Methodologically, results suggest that further research is called for to address the dimensionality problem associated with non-traded factors, building on our missing orthogonal factor approach.

The paper is organized as follows. Section 2 sets the framework. Section 3 provides our inference methodology within a complete model, and Section 4 presents our extension with missing factors. Empirical and simulation results are reported in Section 5. We conclude in Section 6.

2. Model formulation and identification

Linear multi-factor asset pricing models (LAPM) stipulate that the unconditional expectation of returns (μ_r) is linear in factor loadings (or *betas*):

$$\mu_r = \iota_n \gamma_c + b' \Gamma \quad (2.1)$$

where μ_r is an $n \times 1$ vector, $\iota_n = (1, \dots, 1)'$ is an $n \times 1$ vector of ones, γ_c is the *zero-beta* rate (a cross-sectional intercept), $b = [b_1 \dots b_n]$ is a $q \times n$ matrix of factor loadings representing exposures to q factors, and $\Gamma = (\Gamma_1, \dots, \Gamma_q)'$ is a vector of associated premiums; see e.g. Ross (1976), Huberman et al. (1987), Shanken and Zhou (2007), and the review of Campbell et al. (1997, Chapter 6, Equation 6.1.8). All of these parameters are unknown.

The relation (2.1) is typically taken to the data as follows. We suppose we have $T \times 1$ vectors r_i of observations on n assets ($i = 1, \dots, n$) over the period $t = 1, \dots, T$, and $\mathcal{R} = [\mathcal{R}_1 \dots \mathcal{R}_q] = [\mathcal{R}'_1, \dots, \mathcal{R}'_q]'$ a $T \times q$ matrix of observations on a set of q factors which may explain returns, where $\mathcal{R}_j = (\mathcal{R}_{j1}, \dots, \mathcal{R}_{jT})'$ and $\mathcal{R}_{jt}$ is the observation on factor j at time t . It is then typically assumed that $\mathcal{R}_{jt}$ has a time-invariant factor mean $\mu_{\mathcal{R}_j}$:

$$\mu_{\mathcal{R}} = (\mu_{\mathcal{R}_1}, \dots, \mu_{\mathcal{R}_q})', \quad \mu_{\mathcal{R}_j} = E(\mathcal{R}_{jt}), \quad j = 1, \dots, q, \quad t = 1, \dots, T.$$

The empirical counterparts of the above are denoted

$$\bar{\mathcal{R}} = (\bar{\mathcal{R}}_1, \dots, \bar{\mathcal{R}}_q)', \quad \bar{\mathcal{R}}_j = \frac{1}{T} \sum_{t=1}^T \mathcal{R}_{jt}, \quad j = 1, \dots, q.$$

A factor can be *tradable* (e.g., if it represents a portfolio of traded securities, such as a market benchmark), or *non-tradable* (such as a macro-economic variable). It is commonly agreed that candidate models should attempt to price tradable factors (Lewellen et al., 2010, Prescription 4), in the sense that expected returns on a traded factor should itself satisfy (2.1). To describe how to do so, assume that $\mathcal{R}_1$ is tradable so that $\mathcal{R} = [\mathcal{R}_1 \ F]$ where $F = [\mathcal{R}_2 \dots \mathcal{R}_q]$ collects the non-traded factors.³ Then b is introduced through the regression:

$$r_i = a_i \iota_T + \mathcal{R}_1 b_{i1} + F b_{iF} + u_i, \quad i = 1, \dots, n, \quad (2.2)$$

² See Timmermann (2008) for different although related arguments on intermittent predictability.

³ We consider a single traded factor for notational ease; extensions to multiple ones follow straightforwardly.

where $b_i = (b_{i1}, b'_{iF})'$, b_{i1} is a scalar, and b_{iF} is a $(q-1) \times 1$ vector. Ignoring for now that $\mathcal{R}_1$ is tradable, (2.2) can be formulated as follows (Campbell et al., 1997, Chapter 6)⁴:

$$r_i - \gamma_c \iota_T = (\mathcal{R}_1 - \gamma_0 \iota_T) b_{i1} + (F - \iota_T \gamma'_F) b_{iF} + u_i, \quad i = 1, \dots, n, \quad (2.3)$$

$$(\gamma_0, \gamma'_F)' = \mu_{\mathcal{R}} - \Gamma := \theta. \quad (2.4)$$

This means that the intercepts of (2.2) are constrained as follows:

$$a_i = \gamma_c - \gamma_0 b_{i1} - \gamma'_F b_{iF}, \quad i = 1, \dots, n. \quad (2.5)$$

Here, γ_c and γ_0 are free parameters (i.e., $\gamma_c \neq \gamma_0$). Yet, if $\mathcal{R}_1$ satisfies (2.1), then $\Gamma_1 = \mu_{\mathcal{R}_1} - \gamma_c$ which, given (2.4), implies that $\gamma_0 = \gamma_c$. Substituting γ_0 for γ_c in (2.3) amounts to imposing

$$a_i = \gamma_0(1 - b_{i1}) - \gamma'_F b_{iF}, \quad i = 1, \dots, n, \quad (2.6)$$

on (2.2). When all factors are tradable, extending (2.6) corresponds to mean–variance efficiency (Shanken, 1986), (Beaulieu et al., 2013), yielding the restriction

$$a_i = \gamma_0(1 - b_{i1} - \iota'_{(q-1)} b_{iF}), \quad i = 1, \dots, n. \quad (2.7)$$

To interpret these restrictions, consider the special case where all factors are tradable, and a riskless asset with return r_f is available. Then, replacing γ_c , γ_0 and each component of γ_F by r_f in (2.3) gives the zero-intercept Sharpe–Lintner model. Restrictions (2.6) and (2.7) matter for model assessment. When premiums are unrestricted as in (2.5), γ_c holds critical information (Barone-Adesi et al., 2004; Lewellen et al., 2010 and Penaranda and Sentana, 2016). The above alternative model formulations and restrictions motivate our contribution.

Inference in this context involves enduring hurdles, for b is unobserved. It is clear from (2.1) that Γ is not identified unless b has full rank. The same condition is required for θ to be recoverable with no further information. The rank of b may fail for various reasons; special cases raised by Kleibergen (2009) and Beaulieu et al. (2013) relate to β etas jointly (across i) equal to zero or one, where the former case may be traced back to Kan and Zhang (1999). More generally, using portfolios rather than individual assets for r_i tends to equalize β etas across i . Rank failures are not restricted to such drivers (Kleibergen et al., 2021 and Kleibergen and Zhan, 2020) and reflect the information conveyed not just by the factors, but also by the returns r_i considered.

Sorting out the multiple sources of identification failure is difficult. Instead, we propose here confidence sets which preserve the LAPM-based association between all model parameters, including realized or sample dependent random ones. In particular, an alternative parameter [introduced by Shanken, 1985, 1992] as the *ex-post* premium

$$\Gamma_* = \Gamma + \bar{\mathcal{R}} - \mu_{\mathcal{R}} = \bar{\mathcal{R}} - \theta \quad (2.8)$$

has recently regained interest (Khalaf and Schaller, 2016; Jegadeesh et al., 2019, and Kim and Skoulakis, 2018). Γ_* is attractive because it bypasses the unknown $\mu_{\mathcal{R}}$ problem. This problem has lead to two-pass methodologies [as reviewed in e.g. Shanken and Zhou (2007)].⁵ Instead, θ can be estimated even though $\mu_{\mathcal{R}}$ is unknown (Shanken, 1992; Campbell et al., 1997, Chapters 5 & 6). From there on, Γ_* can be estimated conditioning on the factors. The gains from using Γ_* are especially notable in finite samples, since $\bar{\mathcal{R}}$ can deviate markedly from $\mu_{\mathcal{R}}$ in some subperiods. Consequently, we focus on Γ_* from a fundamentally finite-sample perspective.

Restrictions (2.6) or (2.5) are determined by θ , so a change in one of its elements may be “cancelled” by a change in another. We thus derive a confidence region for θ conditional on the factors, and project this region to build simultaneous confidence sets for its components. We then assess pricing reflecting zero-restrictions on the components of $\bar{\mathcal{R}} - \theta$: each factor is considered not priced if its empirical mean is not covered by the confidence set associated with the corresponding component of θ . Decisions on pricing are also simultaneous. Our method also yields a confidence set for γ_c , and a test of the restriction $\gamma_0 = \gamma_c$, conducted via the reparametrization

$$\gamma_c - \gamma_0 b_{i1} - \gamma'_F b_{iF} = \gamma_c^* - \gamma_0 (b_{i1} - 1) - \gamma'_F b_{iF}, \quad \gamma_c^* = \gamma_c - \gamma_0, \quad (2.9)$$

by checking whether the confidence set for γ_c^* covers zero. We thus introduce the hypothesis

$$H_c^* : \gamma_c^* \equiv \gamma_c - \gamma_0 = 0. \quad (2.10)$$

Another special case arises when a factor is a return spread, so its *ex-ante* price should be equal to its expectation (Shanken and Weinstein, 2006). In this case, one may test whether the associated component of γ_F is zero. An alternative formulation [proposed by Kleibergen (2009), Kleibergen et al. (2021) and Kleibergen and Zhan (2020)] involves expressing the individual equations of (2.3) in deviation from one of them, leading to $n-1$ regressions of the form:

$$r_i - r_j = a_i^* + (\mathcal{R}_1 - \gamma_0 \iota_T) b_{i1}^* + (F - \iota_T \gamma'_F) b_{iF}^* + u_i^*, \quad (2.11)$$

⁴ Derivations are as follows: taking unconditional expectations of the unconstrained (2.2) regression with a time invariant perspective implies $\mu_r = (a_1, \dots, a_n)' + b' \mu_{\mathcal{R}}$, which equated with (2.1) yields (2.5).

⁵ “An average return carries no information about the mean of the factor that is not already observed in the sample mean of the factor.” (Cochrane, 2005, p. 245). See also Penaranda and Sentana (2016), on including moment conditions on factor means with GMM.

$$a_i^* = 0, \quad i = 1, \dots, n-1, \quad (2.12)$$

for any given $j \neq i \in \{1, \dots, n\}$. A confidence set can then be obtained for each component of (γ_0, γ_F') , which is invariant to the reference equation. Interpretations on pricing are unchanged, at the expense of foregoing information on γ_c which is evacuated by subtraction from the estimating equations. This implies that H_c^* cannot be assessed nor can mean-variance efficiency. By contrast, our approach considers all the above formulations.

3. Inference with observed factors

The above asset pricing models can be defined via rank restrictions on the multivariate regression:

$$Y = XB + U \quad (3.1)$$

where $Y := [Y_1, \dots, Y_T]'$ is a $T \times n$ matrix of observations Y_t' [the t th row of Y] on n endogenous variables, $X = [X_1, \dots, X_T]'$ is a $T \times k$ full-column rank matrix of exogenous variables, and $U = [U_1, \dots, U_T]'$ is a $T \times n$ matrix of random errors. To obtain finite-sample results, we shall assume that (conditional on X)

$$U_1, \dots, U_T \stackrel{i.i.d.}{\sim} N[0, \Sigma], \det(\Sigma) > 0. \quad (3.2)$$

Throughout the paper, we use the following notation. $s_m[j]$ denotes an $m \times 1$ selection vector, whose all elements are zero, except for the j th (equal to 1). The number of factors is $q = k - 1$. For any full-column rank matrix A , $\mathcal{M}[A] = I - A(A'A)^{-1}A'$. $\text{diag}(A)$ refers to a column vector from the diagonal of a matrix A . For any $N \times K$ matrix A , $\text{vec}(A)$ returns an $(NK) \times 1$ vector, with the columns of A stacked on top of each other. The symbol $\overset{H}{\sim}$ refers to the null distribution of a statistic under a given hypothesis H , $F(n, m)$ represents the Fisher distribution with degrees of freedom (n, m) , and $F_\alpha(n, m)$ is the α -level critical point from the latter distribution. On applying least-squares to (3.1), we get

$$\hat{B} = (X'X)^{-1}X'Y, \quad \hat{S} = \hat{U}'\hat{U}, \quad \hat{U} = Y - X\hat{B}. \quad (3.3)$$

It will also be convenient to consider the following matrix partitions:

$$(X'X)^{-1} = \begin{bmatrix} x^{11} & x^{12} \\ x^{21} & x^{22} \end{bmatrix}, \quad (3.4)$$

$$B = \begin{bmatrix} a' \\ b \end{bmatrix}, \quad \hat{B} = \begin{bmatrix} \hat{a}' \\ \hat{b} \end{bmatrix}, \quad b = [b_1 \quad \dots \quad b_n] = \begin{bmatrix} \beta_2' \\ \vdots \\ \beta_k' \end{bmatrix}, \quad \hat{b} = \begin{bmatrix} \hat{\beta}_2' \\ \vdots \\ \hat{\beta}_k' \end{bmatrix}, \quad (3.5)$$

where x^{11} is a scalar, $x^{21} = x^{12'}$ is $q \times 1$ and x^{22} is $q \times q$, $a = (a_1, \dots, a_n)'$ is the vector of n intercepts, and b is $q \times n$. The rank restrictions of interest can be written as follows:

$$H_R : (1, \theta_R')B = 0, \quad \text{for some unknown vector } \theta_R, \quad (3.6)$$

$$H_U : (1, \theta_U')B = \phi_0', \quad \text{for some unknown vector } \theta_U = (\theta_U', \phi_0)', \quad (3.7)$$

where θ_R and θ_U are $q \times 1$ vectors and ϕ is an unknown scalar.⁶ Indeed, the non-tradable case (2.2) - (2.5) corresponds to (3.1) where

$$X = [\iota_T, F], \quad Y := [r_1, \dots, r_n], \quad (3.8)$$

imposing (3.7) where θ_U corresponds to θ in (2.4), and $\phi := \gamma_c$. On rewriting (2.2) - (2.6) with left-hand side returns in deviation from $\mathcal{R}_1$, we get:

$$r_i - \mathcal{R}_1 = a_i \iota_T + (\mathcal{R}_1 - \iota_T \gamma_0) d_i + (F - \iota_T \gamma_F') b_{iF} + u_i, \quad i = 1, \dots, n, \quad (3.9)$$

where $a_i = 0$, $d_i = b_{i1} - 1$. This regression is a special case of (3.1) - (3.8) under (3.6), with

$$Y := [r_1 - \mathcal{R}_1, \dots, r_n - \mathcal{R}_1], \quad b = [d, b_F']', \quad d = (d_1, \dots, d_n)', \quad b_F = [b_{1F} \quad \dots \quad b_{nF}], \quad (3.10)$$

where θ_R corresponds to θ in (2.4). Regression (2.2) - (2.5) can also be re-expressed as

$$r_i - \mathcal{R}_1 = a_i \iota_T + \mathcal{R}_1 d_i + F b_{iF} + u_i, \quad i = 1, \dots, n, \quad (3.11)$$

in which (3.7) with $\phi = \gamma_c^*$ incorporates (2.9) and thus allows one to test (2.10). This yields again (3.1) - (3.8) with (3.7) and (3.10), where θ_U is θ in (2.4) and $\phi := \gamma_c^*$.

Our objective is to propose confidence sets for θ_R and θ_U whose level is controlled in finite samples without assuming identification [i.e., without an assumption on the rank of B]. For this purpose, we introduce Hotelling-type statistics for special cases of H_R and H_U :

$$H_{0R} : (1, \theta_{R0}')B = 0, \quad \theta_{R0} \text{ known}, \quad (3.12)$$

$$H_{0U} : (1, \theta_{U0}')B = \phi_0', \quad \theta_{U0} = (\theta_{R0}', \phi_0)' \text{ known}. \quad (3.13)$$

⁶ Typically, (3.6) and (3.7) assume that $k \leq n$.

These statistics have null distributions which are invariant to the rank of B . From there on, we propose an analytical solution to invert these statistics for inference on $\theta_{\mathbf{R}}$ and $\vartheta_{\mathbf{U}}$.

Theorem 3.1. Suppose that the assumptions (3.1) - (3.2) hold with (3.8) - (3.10), along with the notations (3.3) - (3.5), let $\theta_{\mathbf{R}}$ and $\vartheta_{\mathbf{U}} = (\theta'_{\mathbf{U}}, \phi)'$ be defined by (3.6) and (3.7). Then, regardless of the rank of B , the confidence set

$$CS_{\alpha}(\vartheta_{\mathbf{U}}) = \{\vartheta_{\mathbf{U}0} : \Omega(\vartheta_{\mathbf{U}0}) \frac{\tau_n}{n} \leq F_{\alpha}(n, \tau_n)\}, \text{ with} \quad (3.14)$$

$$\tau_n = T - k - n + 1, \quad \Omega(\vartheta_{\mathbf{U}0}) = \frac{[(1, \theta'_{\mathbf{U}0})\hat{B} - \phi_0' l'_n] \hat{S}^{-1} [\hat{B}'(1, \theta'_{\mathbf{U}0})' - \phi_0 l'_n]}{(1, \theta'_{\mathbf{U}0})(X'X)^{-1}(1, \theta'_{\mathbf{U}0})'}, \quad (3.15)$$

has level $1 - \alpha$ for $\vartheta_{\mathbf{U}}$, and the confidence set

$$CS_{\alpha}(\theta_{\mathbf{R}}) = \{\theta_{\mathbf{R}0} : \frac{\tau_n \Lambda(\theta_{\mathbf{R}0})}{n} \leq F_{\alpha}(n, \tau_n)\}, \text{ with } \Lambda(\theta_{\mathbf{R}0}) = \frac{(1, \theta'_{\mathbf{R}0})\hat{B}\hat{S}^{-1}\hat{B}'(1, \theta'_{\mathbf{R}0})'}{(1, \theta'_{\mathbf{R}0})(X'X)^{-1}(1, \theta'_{\mathbf{R}0})'}, \quad (3.16)$$

has level $1 - \alpha$ for $\theta_{\mathbf{R}}$.

Proof. Due to the normality assumption (3.2)

$$\Lambda(\theta_{\mathbf{R}0}) \frac{\tau_n}{n} \sim F(n, \tau_n), \quad \Omega(\vartheta_{\mathbf{U}0}) \frac{\tau_n}{n} \sim F(n, \tau_n)$$

when $\theta_{\mathbf{R}0}$ and $\vartheta_{\mathbf{U}0}$ are the true model parameters. The latter distributional results do not require any identification restriction; see Dufour and Khalaf (2002) and references therein.⁷ This yields the desired result for the confidence sets that invert associated tests. ■

The normality assumption underlying Theorem 3.1 provides exact distributional theory, yet it is not central to our results. Asymptotic counterparts will hold under standard regularity assumptions [on (3.1)], which do not require identification. Indeed, the LR-type counterpart of the underlying Hotelling statistics are monotonic transformation of the latter:

$$\mathcal{L}_{\Omega}(\vartheta_{\mathbf{U}0}) = T \ln\left(1 + \Omega(\vartheta_{\mathbf{U}0})\right), \quad \mathcal{L}_{\Lambda}(\theta_{\mathbf{R}0}) = T \ln\left(1 + \Lambda(\theta_{\mathbf{R}0})\right),$$

and will have $\chi^2(n)$ null distributions; see Dufour and Khalaf (2002), or Berndt and Savin (1977) and Gouriéroux et al. (1995).⁸

Consider now the $k \times k$ matrix $A_{\mathbf{R}}$ and the $(k+1) \times (k+1)$ matrix $A_{\mathbf{U}}$ defined as follows:

$$A_{\mathbf{R}} := \hat{B}\hat{S}^{-1}\hat{B}' - \frac{n}{\tau_n}F_{\alpha}(n, \tau_n)(X'X)^{-1}, \quad A_{\mathbf{U}} := \begin{bmatrix} A_{\mathbf{R}} & -\hat{B}\hat{S}^{-1}l'_n \\ -l'_n\hat{S}^{-1}\hat{B}' & l'_n\hat{S}^{-1}l'_n \end{bmatrix}, \quad (3.17)$$

along with the additional partitions

$$A_{\mathbf{R}} = \begin{bmatrix} A_{\mathbf{R}11} & A_{\mathbf{R}12} \\ A'_{\mathbf{R}12} & A_{\mathbf{R}22} \end{bmatrix}, \quad A_{\mathbf{U}} = \begin{bmatrix} A_{\mathbf{U}11} & A_{\mathbf{U}12} \\ A'_{\mathbf{U}12} & A_{\mathbf{U}22} \end{bmatrix} \quad (3.18)$$

where $A_{\mathbf{R}11}$ and $A_{\mathbf{U}11}$ are scalar [i.e., the element (1, 1) of each matrix]. The confidence sets of Theorem 3.1 can then be expressed in closed form.

Corollary 3.1.1. Under the assumptions of Theorem 3.1, the confidence sets $CS_{\alpha}(\vartheta_{\mathbf{U}})$ and $CS_{\alpha}(\theta_{\mathbf{R}})$ can be rewritten as follows:

$$CS_{\alpha}(\vartheta_{\mathbf{U}}) = \{\vartheta_{\mathbf{U}0} : \vartheta'_{\mathbf{U}0}A_{\mathbf{U}22}\vartheta_{\mathbf{U}0} + 2A_{\mathbf{U}12}\vartheta_{\mathbf{U}0} + A_{\mathbf{U}11} \leq 0\}, \quad (3.19)$$

$$CS_{\alpha}(\theta_{\mathbf{R}}) = \{\theta_{\mathbf{R}0} : \theta'_{\mathbf{R}0}A_{\mathbf{R}22}\theta_{\mathbf{R}0} + 2A_{\mathbf{R}12}\theta_{\mathbf{R}0} + A_{\mathbf{R}11} \leq 0\}, \quad (3.20)$$

where $A_{\mathbf{U}11}$, $A_{\mathbf{U}12}$, $A_{\mathbf{U}22}$, $A_{\mathbf{R}11}$, $A_{\mathbf{R}12}$ and $A_{\mathbf{R}22}$ are defined in (3.17) - (3.18).

Proof. Using the definition of $\Lambda(\theta_{\mathbf{R}0})$ in (3.16), $\Lambda(\theta_{\mathbf{R}0}) \tau_n/n \leq F_{\alpha}(n, \tau_n)$ corresponds to

$$\frac{(1, \theta'_{\mathbf{R}0})\hat{B}\hat{S}^{-1}\hat{B}'(1, \theta'_{\mathbf{R}0})' \tau_n}{(1, \theta'_{\mathbf{R}0})(X'X)^{-1}(1, \theta'_{\mathbf{R}0})' n} < F_{\alpha}(n, \tau_n)$$

which can be rewritten as

$$(1, \theta'_{\mathbf{R}0})[\hat{B}\hat{S}^{-1}\hat{B}' - F_{\alpha}(n, \tau_n)(X'X)^{-1}n/\tau_n](1, \theta'_{\mathbf{R}0})' < 0.$$

⁷ Other than the usual Least Squares assumptions on $X'X$ and $\hat{S}$, of course.

⁸ Simulations reported below show that for the problem under consideration corresponding to 5 or 10 year subsamples, the normal cut-off controls size whether errors are multivariate Student- t or in the presence of GARCH effects.

Furthermore, from (3.17) we have $A_R := \hat{B}\hat{S}^{-1}\hat{B}' - F_\alpha(n, \tau_n)(X'X)^{-1}n/\tau_n$ leading to

$$(1, \theta'_{0R})A_R(1, \theta'_{0R})' < 0.$$

Next, applying the partitioning of A_R in (3.18) yields

$$(1, \theta'_{0R})A_R(1, \theta'_{0R})' = A_{R11} + 2A_{R12}\theta_{0U} + \theta'_{0U}A_{R12}\theta_{0U}$$

which proves (3.20). Turning to $\Omega(\theta_{0U})$ as in (3.15), $\Omega(\theta_{0U})\tau_n/n \leq F(n, \tau_n)$ corresponds to

$$\frac{[(1, \theta'_{0U})\hat{B} - \phi_0 t'_n]\hat{S}^{-1}[\hat{B}'(1, \theta'_{0U})' - \phi_0 t_n]}{(1, \theta'_{0U})(X'X)^{-1}(1, \theta'_{0U})'} \frac{\tau_n}{n} < F_\alpha(n, \tau_n)$$

which, using (3.17), can be rewritten as

$$(1, \theta'_{0U})A_R(1, \theta'_{0U})' - 2\phi_0(1, \theta'_{0U})\hat{B}\hat{S}^{-1}t_n + \phi_0^2 t'_n \hat{S}^{-1}t_n < 0.$$

Now introduce the $(q+1) \times 1$ vector $\zeta_0 = (\theta'_{0U}, \phi_0)'$ and consider the inequality

$$(1, \zeta'_0)A_U(1, \zeta'_0)' \leq 0. \quad (3.21)$$

From (3.18) and observing that $\phi_0(1, \theta'_{0U})\hat{B}\hat{S}^{-1}t_n = \phi_0 t'_n \hat{S}^{-1}\hat{B}'(1, \theta'_{0U})'$, we have

$$(1, \zeta'_0)A_U(1, \zeta'_0)' = (1, \theta'_{0U})[A_R](1, \theta'_{0U})' - 2\phi_0(1, \theta'_{0U})\hat{B}\hat{S}^{-1}t_n + \phi_0^2 t'_n \hat{S}^{-1}t_n$$

which entails that inequality $\Omega(\theta_{0U})\tau_n/n \leq F(n, \tau_n)$ corresponds to (3.21). Finally, the partitioning of A_U in (3.18) allows us to expand $(1, \zeta'_0)A_U(1, \zeta'_0)'$ as follows

$$(1, \zeta'_0)A_U(1, \zeta'_0)' = A_{U11} + 2A_{U12}\zeta_0 + \zeta'_0 A_{R12}\zeta_0$$

which proves (3.19). ■

The special case $q = 1$ in (3.19) coincides with the unidimensional quadratic inequality proposed by Beaulieu et al. (2013) for a tradable factor. The general regions we derive here are multi-dimensional *quadratics* sets:

$$\zeta' A_{22}\zeta + 2A_{12}\zeta + A_{11} \leq 0 \quad (3.22)$$

where ζ is the $m \times 1$ vector of unknown parameter, and A_{22} , A_{12} , A_{11} are data dependent matrices [see Dufour and Taamouti (2005)].⁹ Analytical solutions for projections-based CS for any linear transformation of ζ of the form $\omega'\zeta$ can then be obtained, the outcome of which can be empty, bounded, or the union of two unbounded disjoint sets or equal to $\mathbb{R}$.

The condition for empty sets which generalizes the single-beta results in Beaulieu et al. (2013) can thus be stated as follows. Because the cut-off point underlying test inversion denoted $F_\alpha(n, \tau_n)$ above is the same for all parameter values, empty sets would result when $\min_{\theta_{0U}} \Omega(\theta_{0U}) \geq F_\alpha(n, \tau_n)$ or $\min_{\theta_{0R}} \Lambda(\theta_{0R}) \geq F_\alpha(n, \tau_n)$. It can be shown that minimizing these statistics in turn produces the Gaussian-LR statistic to test the nonlinear restriction which defines the underlying parameters, namely (3.6) and (3.7); general derivation are available in e.g. Gouriéroux et al. (1996). We next give a simple sufficient but not necessary condition under which such confidence sets are unbounded which indicates identification failure.

Theorem 3.2. *Under the assumptions of Theorem Theorem 3.1, consider the Hotelling statistics*

$$A_{0i} = \frac{s_k[i]'\hat{B}\hat{S}^{-1}\hat{B}'s_k[i]}{s_k[i]'(X'X)^{-1}s_k[i]}, \quad i \in \{2, \dots, k\}$$

associated with the hypotheses

$$H_{0i} : s_k[i]'B = 0, \quad i \in \{2, \dots, k\}.$$

If

$$(\tau_n/n)A_{0i} < F_\alpha(n, \tau_n), \quad i \in \{2, \dots, k\}$$

then the confidence sets for θ_R and for $\theta_U = (\theta'_{0U}, \phi_0)'$ as in (3.16) and (3.14) are unbounded.

Proof. From Corollary 3.1.1, the confidence sets defined by (3.14) and (3.16) take the quadrics form (3.22). For sets of this form, Dufour and Taamouti (2005) show that the confidence sets are unbounded if A_{22} is not positive definite. For the set defined by (3.16), using the partitionings (3.4) and (3.17), we have:

$$A_{R11} = \hat{a}'\hat{S}^{-1}\hat{a} - ((n/\tau_n)F_\alpha(n, \tau_n))x^{11}, \quad A_{22R} = \hat{b}\hat{S}^{-1}\hat{b}' - [(n/\tau_n)F_\alpha(n, \tau_n)]x^{22}, \\ A_{12R} = A'_{21R} = \hat{a}'\hat{S}^{-1}\hat{b}' - [(n/\tau_n)F_\alpha(n, \tau_n)]x^{12}.$$

⁹ Dufour and Taamouti (2005) consider the inversion of an altogether different test statistic for an altogether different econometric model, namely the single instrumental variable regression equation.

The diagonal term of A_{22R} is thus given by $\text{diag}(A_{22}) = [F_2 \dots F_k]'$ where

$$F_i = s_k[i]' \hat{B} \hat{S}^{-1} \hat{B}' s_k[i] - s_k[i]' (X'X)^{-1} s_k[i] \frac{n f_{n, \tau_n, \alpha}}{\tau_n}.$$

Clearly, if any of the Hotelling tests based on Λ_{0i} , $i \in \{2, \dots, k\}$ is not significant at level α relative to the null distribution $\Lambda_{0j} \stackrel{H_{0j}}{\sim} F(n, \tau_n)$, then by the definition of Λ_{0i} and F_i ,

$$\tau_n \Lambda_{0i} / n < F_\alpha(n, \tau_n) \Leftrightarrow F_i < 0$$

in which case A_{22R} cannot be positive definite. By comparing A_R and A_U in (3.17), we see that the same condition also holds for the confidence set defined in (3.14). ■

This theorem shows that if any of the non-traded factors is redundant from a joint significance perspective, then information on prices for all factors is compromised. The same holds if the betas on any traded factor are jointly equal to one. These conditions are sufficient but not necessary, because the rank condition underlying identification can fail for other reasons. It follows that although Hotelling tests on each factor are useful, they remain insufficient, and perhaps more importantly, are embedded in our methodology without compounding type-I errors.

To conclude, we introduce two further tests, whose inversion can be implemented as in Theorem 3.1. The first one is based on a statistic that has the same form and the same null distribution as $\Omega(\theta_{U0})$, yet is applied to (3.11) for inference on γ_c^* as defined in (2.9). This parameter serves to assess the traded factor assumption, namely (2.10), which holds crucial information about the specification. While γ_c^* may be viewed as a reparametrization with respect to γ_c , invariance to such transformations should not be taken for granted. In the context of (3.1) where Y and X are given by (3.8) and (3.10), $\theta_{U0} = (\theta'_{0U}, \gamma_{0c}^*)'$ where θ_{0U} and γ_{0c}^* are known, we thus propose the following statistic:

$$\Omega_*(\theta_{U0}) = \frac{[(1, \theta'_{0U}) \hat{B} - \gamma_{0c}^* \hat{S}^{-1} \hat{B}' (1, \theta'_{0U})' - \gamma_{0c}^* t_n]}{(1, \theta'_{0U})(X'X)^{-1}(1, \theta'_{0U})'} \quad (3.23)$$

The second test is based on the transformed regression introduced by Kleibergen (2009), Kleibergen et al. (2021) and Kleibergen and Zhan (2020), specifically (2.11)–(2.12). The latter can be re-written in the (3.1) form where $Y := [r_1, \dots, r_n]P$, X is given by (3.8) and P is the associated $n \times (n-1)$ transformation matrix. In this context, we consider the statistic

$$A_P(\theta_{R0}) = \frac{(1, \theta'_{R0}) \hat{B} P(P' \hat{S} P)^{-1} P' \hat{B}' (1, \theta'_{R0})'}{(1, \theta'_{R0})(X'X)^{-1}(1, \theta'_{R0})'} \stackrel{H_{0R}}{\sim} F(n-1, T-q-n+1) \quad (3.24)$$

which is invariant to the reference equation. This statistic closely resembles the GRS-FAR criterion from Kleibergen and Zhan (2020) (equations 23 and 24); indeed, the latter which is designed to be inverted over the ex-ante zero-beta rate Γ , is nevertheless expressed in terms of demeaned factors, which is in deviation from μ_R where $\bar{R}$ is used as its “sample counterpart”. In view of (2.8), empirical demeaning provides finite-sample inference on Γ_* . So to be clear, our comparative simulation and empirical analyses below rely on (3.24).

4. Inference methods with missing factors

In this section we relax the assumption that the above models include all relevant factors. Missing factors in asset pricing are a long-standing concern. Early references include MacKinlay (1995), who introduced deviations from the CAPM that remain compatible with “risk-based” misspecifications; see also MacKinlay and Pastor (2000) and more recently, Stambaugh and Yuan (2017). Empirically, missing factors have led to the plethora of available candidate models. By contrast, we propose to generalize (2.3) to allow for an unobservable factor that completes the LAPM as in Stambaugh and Yuan (2017), but with no further data nor prior information. For generality, the following presentation relaxes tradability restrictions, yet the same approach applies when some or all factors are imposed as tradable.

Formally, we modify (2.3) as follows: in addition to $\mathcal{R}_1$ and $\mathcal{F}$, there exists a factor with returns denoted as the $T \times 1$ vector Q which annuls the intercept and orthogonalizes the error term when added into the regression. Furthermore, Q is orthogonal to the observed factors, which preserves b_{i1} and b_{iF} . This can be framed into the following regression:

$$r_i - \gamma_c t_T = (\mathcal{R}_1 - \gamma_0 t_T) b_{i1} + (F - t_T \gamma_F') b_{iF} + (Q - t_T \gamma_Q) b_{iQ} + u_i, \quad i = 1, \dots, n, \quad (4.1)$$

where b_{i1} and b_{iF} are unchanged relative to (2.3) and, additionally, as in Stambaugh and Yuan (2017), the error covariance matrix Σ takes the $\sigma^2 I_n$ form. In contrast to Stambaugh and Yuan (2017), we do not construct Q nor introduce restrictions between the weights underlying Q and the intercept, as in MacKinlay and Pastor (2000). We also refrain from adding any further priors as in MacKinlay (1995). Instead, we modify our finite-sample theory to accommodate an unobserved Q .

Regression (4.1) still satisfies (2.1) and is thus considered to be well specified. Of course, Q is not observed so one can only regress returns on $\mathcal{R}_1$ and $\mathcal{F}$. We next show that nevertheless, our inference method on $\theta_U = (\gamma_0, \gamma_F')'$ and γ_{0c} as described in Theorem 3.1 remains valid: the resulting sets will be conservative, in the sense that the underlying tests have level $\leq \alpha$. Since Q is unobserved

and is treated with no further data or modelling inputs, (4.1) is quite general which provides a novel perspective to build on the orthogonal portfolio from MacKinlay (1995).

The above leads to the general regression

$$y = x\beta + \eta + u \quad (4.2)$$

where $y = \text{vec}(Y)$, $x = I_n \otimes X$, $\beta = \text{vec}(B)$, $u = \text{vec}(U)$ as in (3.1), and η is the $(nT) \times 1$ unobserved error component; in (4.1)

$$\eta = (I_n \otimes (Q - \iota_T \gamma_Q))(b_{1Q}, \dots, b_{nQ})'.$$

Hypothesis (3.7) and its linear counterpart (3.13) can be rewritten in this context as

$$\begin{aligned} H_U &: [I_n \otimes (1, \theta_U')] \beta = \gamma_{0c} \iota_n, \text{ for some } \theta_{0U} \text{ and } \gamma_c^*, \\ H_{0U} &: [I_n \otimes (1, \theta_{0U}')] \beta = \gamma_{0c}^* \iota_n, \text{ where } \theta_{0U} \text{ and } \gamma_{0c}^* \text{ are known.} \end{aligned}$$

Then for inversion purposes, we can consider the test based on $\Omega(\vartheta_{U0})$ where $\vartheta_{U0} = (\theta_{0U}', \gamma_{0c}^*)'$ as in Theorem 3.1, here setting $\hat{S}$ to $\hat{\sigma}^2 I_n$, where

$$\hat{\sigma}^2 = (y - x\hat{\beta})'(y - x\hat{\beta}), \quad \hat{\beta} = \text{vec}(\hat{B}).$$

It is straightforward to verify that thus constrained, $\Omega(\vartheta_{U0})$ leads to the usual F-statistic in the regression of y on x , which we denote as $F(\vartheta_{U0})$, up to degree-of-freedom corrections:

$$F(\vartheta_{U0}) = \frac{n(T-k)}{n} \Omega(\vartheta_{U0}).$$

If the model was incomplete, the null distribution of $F(\vartheta_{U0})$ will not correspond to $F[n, n(T-k)]$ because of the unobserved component η in (4.2). However, conditional on η , it then follows from the dominance results of Dufour and Nguyen (2021) for F tests in linear models with missing variables that:

$$\mathbb{P}[F(\vartheta_{U0})] \geq F_\alpha[n, n(T-k)] \leq \mathbb{P}[\bar{F}(\vartheta_{U0})] \geq F_\alpha(n, n(T-k)) = \alpha \quad (4.3)$$

where

$$\begin{aligned} \bar{F}(\vartheta_{U0}) &= \frac{(\varphi\hat{\beta} - \xi)' [\varphi(x'x)^{-1}\varphi']^{-1} (\varphi\hat{\beta} - \xi)/n}{u' M(x) u / n(T-k)}, \\ \varphi &= (I_n \otimes (1, \theta_{0U}')), \quad \xi = \gamma_{0c}^* \iota_n, \quad M(x) = I_T - x(x'x)^{-1}x'. \end{aligned}$$

In other words, the critical region $\bar{F}(\vartheta_{U0}) \geq F_\alpha(n, n(T-k))$ is conservative at level α , hence still provides a valid test.

With reference to (the above cited) pseudo-true value based alternatives, inference based on $F(\vartheta_{U0})$ with (4.3) eschews the need to relax the exact LAPM structure; see MacKinlay and Pastor (2000) on the advantages of avoiding this perspective. MacKinlay and Pastor (2000) also document the superiority of an identity error covariance matrix in the definition of the missing factor (here Q), which has led (Stambaugh and Yuan, 2017) to adopt this definition. Motivated by these arguments, our bound provides an innovative and natural way to maintain the LAPM parameters and pricing structure, which as pointed out by MacKinlay and Pastor (2000), provides “a positive approach” to multifactor asset pricing models “despite their apparent imperfections”. Our bound further suggests insightful directions for practical work. For example, one may intentionally leave out some observable factors while allowing an unobservable one to complete the specification.¹⁰ This is clearly not a general prescription, although related arguments can be found in published work. These include the explicit reluctance expressed by Fama and French (2018) to include momentum, or a suggestion to exclude the book-to-market factor by Fama and French (2015). In a literature where identification problems are becoming more and more evident empirically, such directions can be interpreted soundly through our approach that preserves the parameters of interest.

5. Application to baseline factors

Our analysis begins with a model that includes, in addition to the market factor, the non-tradable liquidity factor (Pástor and Stambaugh, 2003) and an oil price factor (Chen et al., 1986; Ferson and Harvey, 1993; Shanken and Weinstein, 2006), with and without momentum (Carhart, 1997). We aim, through this choice of factors, to bring new insights into the macro-factor literature relative to available identification-robust results that have so far focused on consumption-based factors (Kleibergen and Zhan, 2020). We next revisit the Fama–French three-factor model (Fama and French, 1992, 2018).

The key difference between the considered models is that the liquidity and oil factors in the first one are non-traded. An important question is whether traded factors capture factor exposures more informatively. We take this question to the data through the Fama–French model. With reference to Kleibergen and Zhan (2020) who consider this model from an identification-robust perspective, we note that their methodology (as explained above) evacuates the parameter that allows one to impose or test the tradable factor restriction. These restrictions, when relevant, have information-reduction potential that may facilitate identification; see Huberman and Kandel (1987). Our aim here is to assess this potential empirically. In addition, the Fama–French factors are return spreads,

¹⁰ Our empirical results discussed below may be interpreted along these lines.

which entails alternative restrictions as introduced by [Shanken and Weinstein \(2006\)](#). These authors find that such restrictions have no incidence on standard inference. Here we revisit this evidence using identification-robust methods. Overall, we aim to assess the identifying properties of model-consistent restrictions that reflect the nature of the considered factors. In addition, by comparing tests that assume complete versus incomplete models, we can document the identification versus misspecification trade-offs associated with such restrictions, and illustrate the usefulness of our proposed missing factor approach in managing such trade-offs empirically.

As test assets, we use industry portfolios for the US. The data consists of monthly returns from 1961 to 2010, on 12 portfolios of NYSE, AMEX and NASDAQ firms grouped by standard two-digit industrial classification (SIC). The sectors studied include (1) Consumer nondurables; (2) Consumer durables; (3) Manufacturing; (4) Energy; (5) Chemicals; (6) Business equipment; (7) Telecom; (8) Utilities; (9) Shops; (10) Health; (11) Money; (12) Others. For details on the SIC code, refer to Kenneth French web site from which the data were extracted. We rely on 12 portfolios for power considerations. Fama and French benchmark market, size and book-to-market factors (denoted **MKT**, **SMB** and **HML**) are extracted from Kenneth French web site where the details regarding their construction can be found. The momentum factor (denoted **MOM**) is from the same source. Nontraded factors include nontraded liquidity (INNOV LIQ, denoted below as **IAL**), extracted from Robert Stambaugh's website and the change in the monthly average US dollar price per barrel of crude oil (denoted **SCOP**), extracted from FRED.¹¹

Relative to our notation in Section 2, the traded factor $\mathcal{R}_1 := \text{MKT}$. We refer to the components of θ [as defined in (2.4)] as follows: $\gamma_0 := \theta_{\text{MKT}}$, and then in turn as relevant, θ_j where the subscript is the acronym we used for each factor. We thus provide inference on the cross-sectional intercept $\gamma_c^* = \gamma_c - \theta_{\text{MKT}}$, unless we indicate that it is set to zero. Any further restrictions on these coefficients is indicated when relevant. All confidence sets are derived at the 5% level.

To prepare the ground for our empirical analysis, we first conduct a simulation experiment based on the considered data set. This exercise aims to document the implications of imposing and relaxing (2.10), which will position our proposed tests relative to existing works. Simulation results offer useful insights that substantiate our empirical analysis, which we present next.

5.1. Simulation study

We consider an empirically relevant design based on the above sample and the Fama–French three-factor model in which returns are generated by (2.2) with (2.6) or (2.5). We calibrate designs using observed data on portfolios covering the full sample [$n = 12$ and $T = 624$], and the last sub-period [$n = 12$, $T = 120$].

The factors are drawn as normal with means and variance/covariance matrices calibrated to match the considered sample. We set factor loadings and variance/covariance of disturbances to their OLS estimated counterparts for the observed sample, and compute the cross-sectional two-pass OLS estimates of the zero-beta rate and premiums [denoted $\bar{\theta}_0$], and companion standard errors [denoted $SE(\bar{\theta}_0)$]. We use these cross-sectional estimates to calculate θ as in [Shanken and Zhou \(2007\)](#) and from there on, to initialize the simulations underlying the size study. To assess power, we set the parameter under the alternative as

$$\bar{\theta} = \bar{\theta}_0 + \text{step} \times SE(\bar{\theta}_0)$$

where step measures departure from the null hypotheses; γ_c is calibrated in the same way.¹² The disturbances W_t are generated, in turn, as *i.i.d.* normal, multivariate Student with 5 degrees of freedom, and multivariate GARCH whereby the conditional variance of U_t is given by

$$\Sigma_t = (1 - \mu_1 - \mu_2)\Sigma' + \mu_1 U_{t-1} U_{t-1}' J' + \mu_2 \Sigma_{t-1}$$

as in [Engle and Kroner \(1995\)](#). We use $(\mu_1, \mu_2) = (.15, .80)$. The considered inference methods are *not* corrected for departures from the *i.i.d.* assumption nor from normality. Tests and confidence intervals in what follows are at the 5% and 95% level.

We report empirical rejections over 10000 replications for each parameter. These correspond to tests which are associated with the considered confidence sets: if the 95% confidence set does not cover the null value or is empty, the reported test is considered significant. The proportion of empty confidence sets does not exceed 1.26% in designs which are not misspecified; the case that stands out informatively in this regard is reported in [Table 1](#). In each case, we report: (i) under the heading “True Model”, the specification used to generate data, (ii) under the heading “Empirical Model”, the specification which was considered for estimation and inference, and (iii) a reference to the procedure underlying the considered confidence region. The true and empirical specifications differ only regarding the treatment of $\mathcal{R}_1$. In all figures, the parameters corresponding to the null hypothesis are identified via a dashed vertical line.

[Fig. 1](#) depicts rejection frequencies for inference on θ_{MKT} and the cross-sectional intercepts when relevant. The upper left panel corresponds to our test that imposes (2.10), the upper right panel corresponds to the partialled-out test ([Kleibergen and Zhan, 2020](#)), whereas the lower panel relies on our test which relaxes (2.10) yet provides inference on the intercept. Results reveal the following. First, the power loss that stems from ignoring restriction (2.10), when it holds, is substantial. Second, our unrestricted test and the existing partialled-out one perform almost exactly the same for inference on θ_{MKT} ; partialling-out thus does not improve power. Third, our test on the intercept has excellent properties despite our simultaneous approach: whereas partialling-out evacuates this parameter, our test provides excellent power on this fundamental coefficient without sacrificing any information on θ_{MKT} .

¹¹ Federal Reserve Bank of St. Louis spot crude oil price: West Texas intermediate (WTI) (STISPLC) retrieved from FRED January 17th, 2023.

¹² The cross-sectional OLS estimates that we used to calibrate our design are popular despite their imperfections. We have thus initialized our data generating processes using well understood metrics. For space considerations, the simulation values are not reported here but are available from the authors upon request.

$n = 12, T = 624$	$r_i = a_i \iota_T + \mathcal{R}_1 b_{i1} + \mathcal{F} b_{i\mathcal{F}} + u_i$
True model	$a_i = \gamma_0(1 - b_{i1}) - \gamma'_{\mathcal{F}} b_{i\mathcal{F}}, \quad \gamma_0 = \theta_{\text{MKT}}, \quad \gamma_{\mathcal{F}} = (\theta_{\text{SMB}}, \theta_{\text{HML}})'$
Empirical model	upper left panel $a_i = \gamma_0(1 - b_{i1}) - \gamma'_{\mathcal{F}} b_{i\mathcal{F}}$ upper right panel, lower panel $a_i = \gamma_c - \gamma_0 b_{i1} - \gamma'_{\mathcal{F}} b_{i\mathcal{F}}$
Confidence regions:	upper left panel, based on (3.16); upper right panel, based on (3.24); lower panel, based on (3.15).

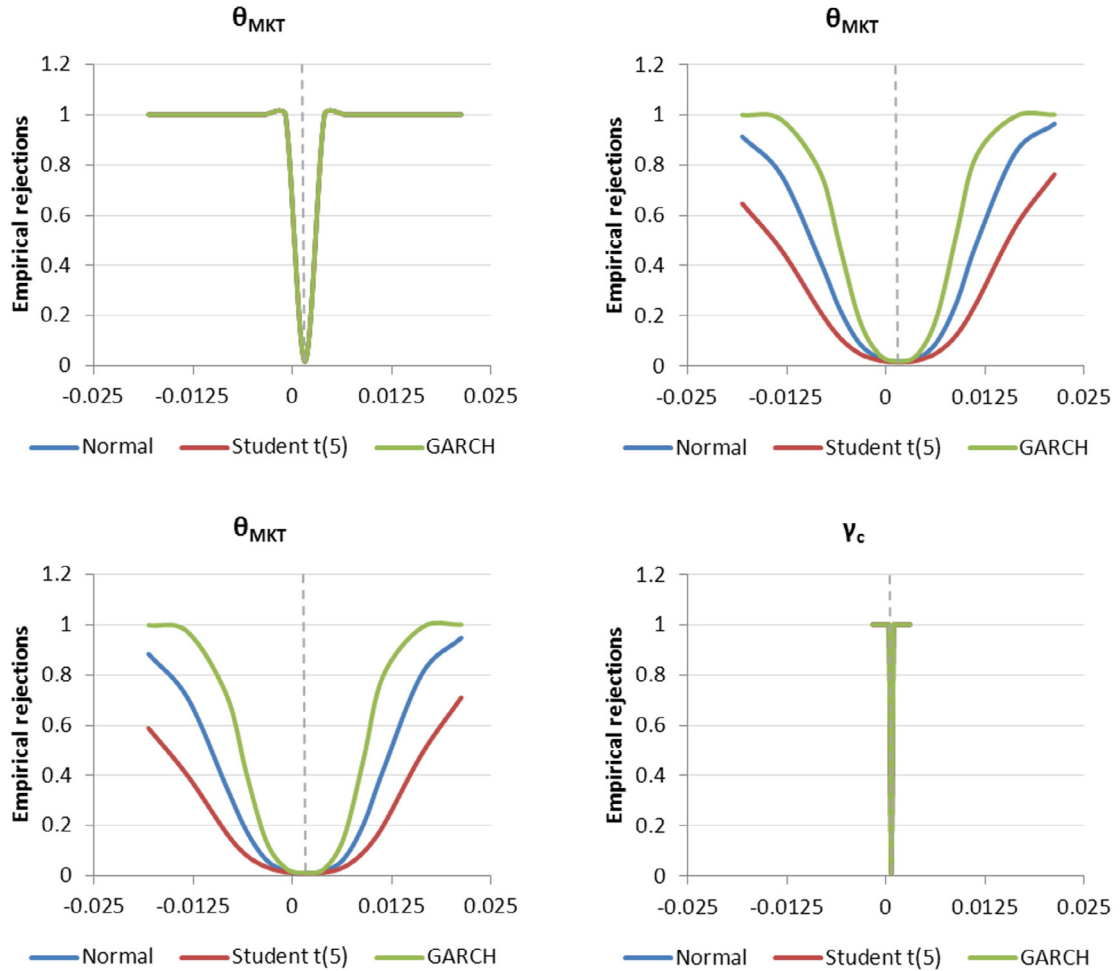


Fig. 1. Power implications of model consistent restrictions.

Note — Dashed vertical lines denotes null parameter values. Numbers reported are empirical rejections pertaining to 5% tests which are associated with the 95% confidence sets: if the 95% confidence set does not cover the null value or is empty, the reported test is considered significant. Parameters are calibrated to OLS cross-sectional two-pass estimates from a training sample based on industry portfolios, Fama–French factors and monthly data, 1961–2010 hence $T = 624$. When one unicolour curve is depicted whereas the legend refers to three cases, this implies that all visually coincide.

We next study the case where (2.10) is imposed when this restriction does not hold. Recall that $\gamma_c \neq \gamma_0$ reflects model fit issues beyond implications for traded factors. The tests we consider pertain to the hypothesis that fixes the value of each component of θ , so in the designs under the null and alternative hypotheses the model will be misspecified because the considered tests falsely fix γ_c to γ_0 . Results are reported in Table 1. We find that empty confidence sets for inference on factor prices occur with very high frequency, even when the data is generated with factor prices which are close to their tested value.

$n = 12, T = 624$	$r_i = a_i \iota_T + \mathcal{R}_1 b_{i1} + \mathcal{F} b_{i\mathcal{F}} + u_i$
True model	$a_i = \gamma_c - \gamma_0 b_{i1} - \gamma'_{\mathcal{F}} b_{i\mathcal{F}}, \quad \gamma_0 = \theta_{\text{MKT}}, \quad \gamma_{\mathcal{F}} = (\theta_{\text{SMB}}, \theta_{\text{HML}})'$
Empirical model	$a_i = \gamma_c - \gamma_0 b_{i1} - \gamma'_{\mathcal{F}} b_{i\mathcal{F}}$
Confidence regions	based on (3.15).

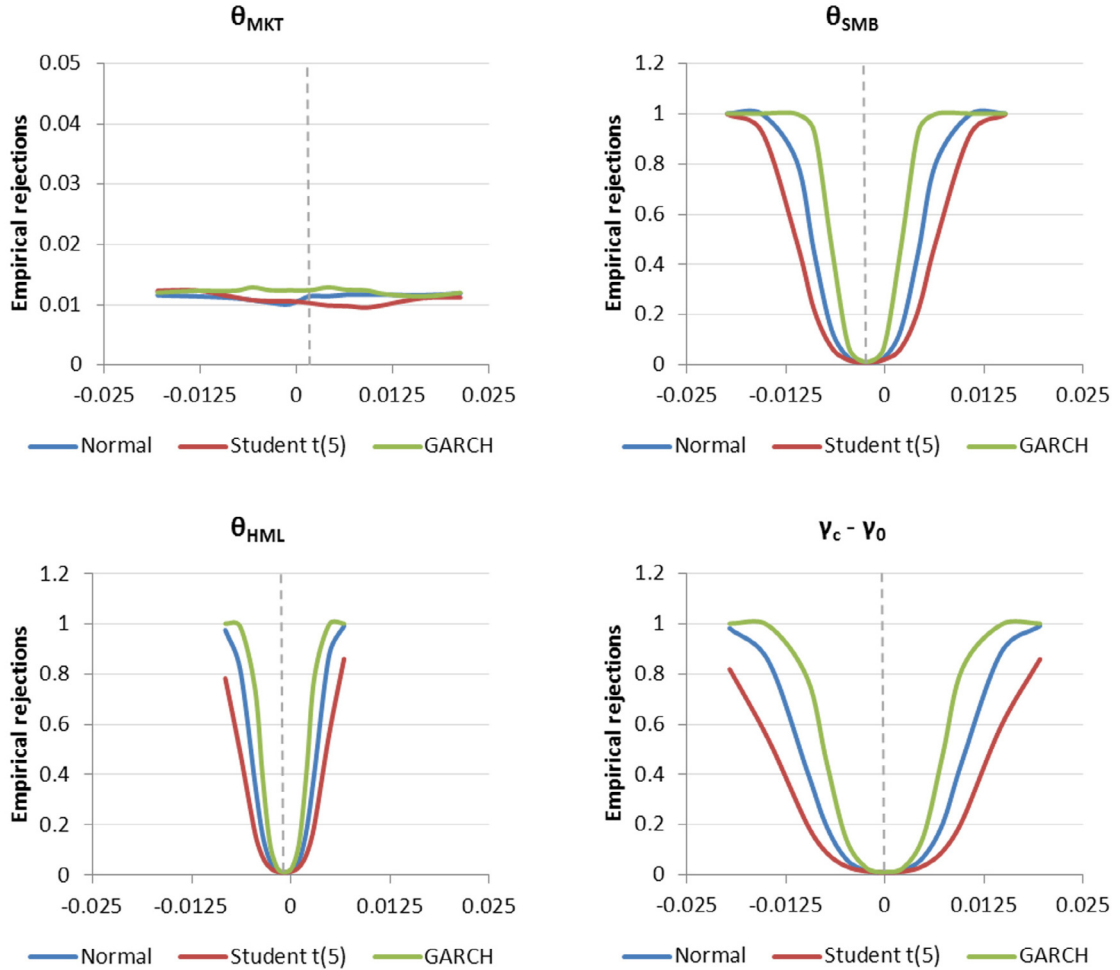


Fig. 2. Information content of the cross-sectional intercept.

Note — See notes to Fig. 1. Under the null hypothesis, we hold $\gamma_c = \gamma_0$, and under the alternative, γ_c moves away from the true value, whereas γ_0 does not. The rejection frequencies in the upper left panel thus depict size (not power).

To relate empty sets to the information content of our test on γ_c^* , we consider a design in which the null hypothesis corresponds to $\gamma_0 = \theta_{\text{MKT}} = \gamma_c$ i.e., $\gamma_c^* = 0$ whereas departures from this hypotheses vary γ_c keeping γ_0 constant. In contrast to Table 1 which relies on our test that imposes (2.10), here we consider instead the test which relaxes this restriction yet provides inference on the intercept. Results are reported in Fig. 2. We see from the upper left panel that size is well controlled for inference on θ_{MKT} . Our tests further display good power on non-market factor prices and, importantly, on γ_c^* . Finally, we provide in Fig. 3 a power study to compare power for sample sizes corresponding to our subperiods, relative to the full sample. Comparing figures across the upper and lower panel, we see that the considered test (here, which imposes (2.10)) has good power even when T drops from 624 to 120. Strikingly, power curves do not differ much between the large and small samples.

To conclude, our simulation results illustrate the power advantages of our proposed procedures. Furthermore, we find that deviations from normality are not distortive in the following sense: no over-rejections occur under the null hypothesis when the *i.i.d.* normal assumption is violated. This result is noteworthy given the prevalence of multivariate GARCH or Student-*t* type assumptions on disturbances in theoretical and empirical asset pricing work.

Upper panel	$n = 12, T = 624$	$r_i = a_i \iota_T + \mathcal{R}_1 b_{i1} + \mathcal{F} b_{i\mathcal{F}} + u_i$
Lower panel	$n = 12, T = 120$	
True model		$a_i = \gamma_0(1 - b_{i1}) - \gamma'_{\mathcal{F}} b_{i\mathcal{F}}, \quad \gamma_0 = \theta_{\text{MKT}}, \quad \gamma_{\mathcal{F}} = (\theta_{\text{SMB}}, \theta_{\text{HML}})'$
Empirical model		$a_i = \gamma_0(1 - b_{i1}) - \gamma'_{\mathcal{F}} b_{i\mathcal{F}}$
Confidence regions: based on (3.16)		

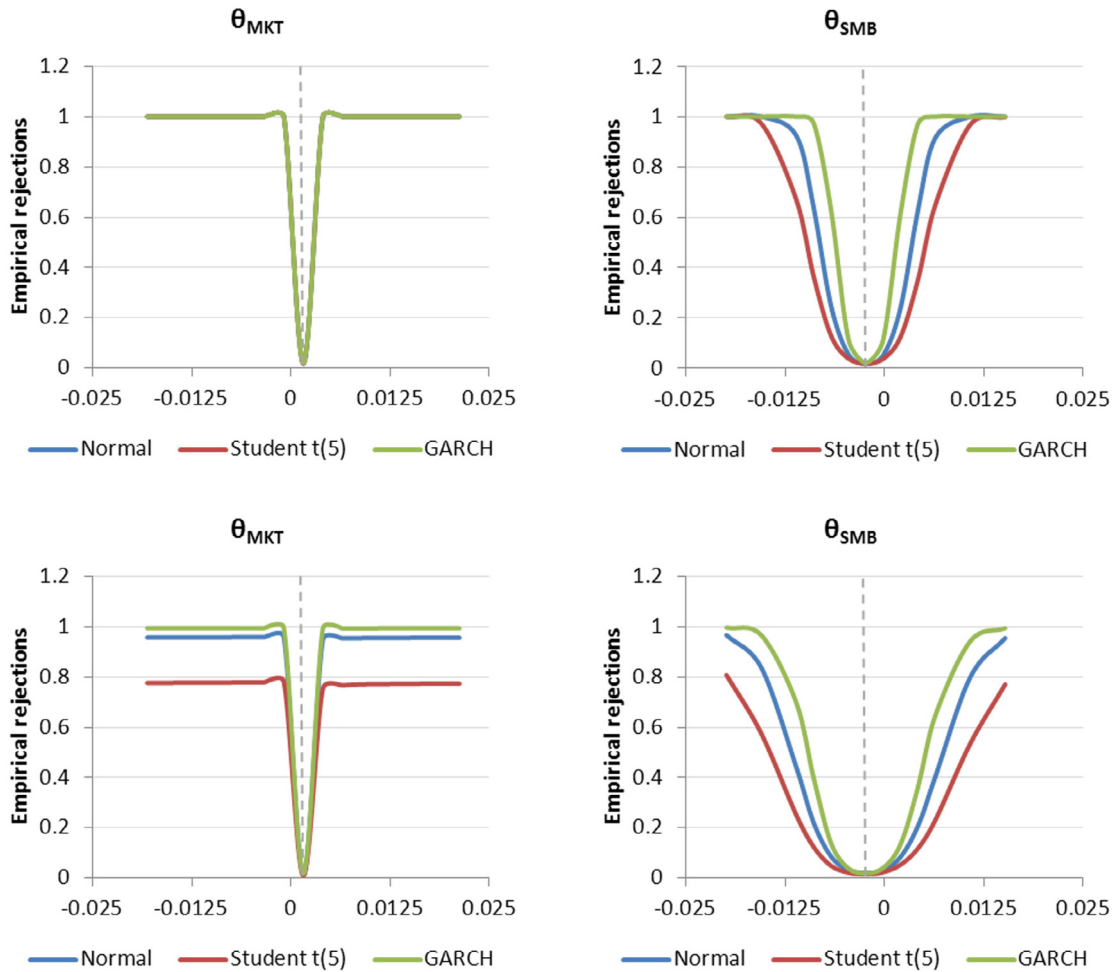


Fig. 3. Effect of sample size. Note — See notes to Fig. 1.

5.2. Empirical results

We now turn to the empirical analysis with the above models and data.

All confidence sets, which we derive at the 5% level, are reported in Panel A of Tables 2–13 assuming complete specifications; results in Panel B allow for missing factors. In all tables, we interpret: (i) the zero cross-sectional intercept test, which checks whether the confidence set on the cross-sectional intercept γ_c^* [as defined in (2.10)] covers zero, and (ii) tests of zero ex-post factor pricing; these check whether the confidence set on each component of θ [as defined in (2.4)] covers the factor empirical mean which we report in each table under the heading $\bar{F}$. The following key points are worth emphasizing to interpret our results. First, because parameters are not identified over the full parameter space, wide and unbounded confidence sets are likely (Dufour, 1997); this does not imply that such outcomes will not inform on some hypotheses of interest, including the ones under consideration. Second, rejections of the hypothesis $\gamma_c^* = 0$ have to be interpreted with caution, particularly in Panel A where tests take a stand on model fit. Third, on tests in Panel B, recall that our distributional results with missing factors yield conservative inference, in the sense that rejections are compelling.

Table 1

Empty confidence sets with misspecified intercept.

$n = 12, \ T = 624$		$r_i = a_i t_T + \mathcal{R}_1 b_{i1} + F b_{iF} + u_i$								
True model		$a_i = \gamma_c - \gamma_0 b_{i1} - \gamma_F' b_{iF}, \quad \gamma_0 = \theta_{\text{MKT}}, \quad \gamma_F = (\theta_{\text{SMB}}, \ \theta_{\text{HML}})'$								
Empirical model		$a_i = \gamma_0(1 - b_{i1}) - \gamma_F' b_{iF}$.								
Confidence regions		based on (3.16).								
		step								
Errors		-4	-3	-2	-1	0	1	2	3	4
Normal		1.0	1.0	1.0	.9700	.6683	.9954	1.0	1.0	1.0
Student t(5)		1.0	1.0	1.0	.9553	.5854	.9930	1.0	1.0	1.0
GARCH		1.0	1.0	1.0	.9811	.7690	.9973	1.0	1.0	1.0

Note — Numbers reported are the proportion of empty confidence sets.

Table 2

Liquidity and oil risk factors.

$r_i - \mathcal{R}_1 = \gamma_c^* t_T + (\mathcal{R}_1 - \gamma_0 t_T) d_i + (F - t_T \gamma'_F) b_{iF} + u_i, \quad i = 1, \dots, n$									
$\gamma_c^* = \gamma_c - \gamma_0, \quad \theta = (\gamma_0, \gamma'_F) = (\theta_{\text{MKT}}, \theta_{\text{IAL}}, \theta_{\text{SCOP}})$									
PANEL A: inverting Hotelling F-test									
CTE		$\mathcal{R}_1$		F		θ_{IAL}		θ_{SCOP}	
$\times 10^{-4}$	γ_c^*	$\overline{F}$	θ_{MKT}	$\overline{F}$	θ_{IAL}	$\overline{F}$	θ_{SCOP}		
61–70	–	38	–	–	–	–	–		
71–80	$\mathbb{R}$	30	$\mathbb{R}$	–124	$\mathbb{R}$	195	$\mathbb{R}$		
81–90	[87, 171]	44	[–298, –131]	13	[225, 413]	–25	[–42, –12]		
91–00	$]-\infty, -330]$ $\cup [2626, \infty[$	103	$]-\infty, -3566]$ $\cup [497, \infty[$	3	$]-\infty, -3124]$ $\cup [-137, \infty[$	3	$]-\infty, -1290]$ $\cup [223, \infty[$		
01–10	$\mathbb{R}$	14	$\mathbb{R}$	–26	$\mathbb{R}$	95	$\mathbb{R}$		
All	$\mathbb{R}$	46	$\mathbb{R}$	–41	$\mathbb{R}$	67	$\mathbb{R}$		
PANEL B: inverting mispricing robust F-test									
61–70	–	38	–	–	–	–	–		
71–80	$\mathbb{R}$	30	$\mathbb{R}$	–124	$\mathbb{R}$	195	$\mathbb{R}$		
81–90	$\mathbb{R}$	44	$\mathbb{R}$	13	$\mathbb{R}$	–25	$\mathbb{R}$		
91–00	$\mathbb{R}$	103	$]-\infty, -3226]$ $\cup [-1268, \infty[$	3	$\mathbb{R}$	3	$\mathbb{R}$		
01–10	$\mathbb{R}$	14	$\mathbb{R}$	–26	$\mathbb{R}$	95	$\mathbb{R}$		
All	$\mathbb{R}$	46	$\mathbb{R}$	–41	$\mathbb{R}$	67	$\mathbb{R}$		

Note — Confidence sets are based on (3.14)–(3.15) and the statistic (3.23), in their Hotelling complete model form in Panel A (see Section 3), and their missing factor counterpart in Panel B (see Section 4). Numbers in bold under the heading $\overline{F}$ indicate factor means that are not covered by the confidence set for the associated component of θ .

Consider first Table 2 which reports our results with the market factor and the non-traded ones, liquidity and oil price. Assuming a complete specification (Panel A), we find that the market factor is priced in 1981–90 and 1991–2000, liquidity is priced in 1981–90 and oil price in 1991–2000. No other effect seems identified outside these decades, nor over the full sample.

The significance of oil price in the 90s and that of liquidity in the 80s contrast sharply with their under-identification in other subperiods and over the full sample. Interestingly, crude oil futures first began trading in 1983. Our results thus underscore the pricing influence of such a radical structural change, albeit with a delayed adjustment into the subsequent decade. As for liquidity, useful insights may emerge from the temporal evolution of the liquidity factor; see Pástor and Stambaugh (2019) (Figure 1). The largest negative realization of liquidity is recorded in 1987, which coincides with a historical financial crisis. This is potentially important in sorting out the identifying role of crises on the factor's pricing influence. However, the 2008 crisis does not seem to play the same role since we obtain real sets as outcomes in that subperiod. On revisiting Figure 1 in Pástor and Stambaugh (2019), we find that the 1987 realization of the liquidity factor stands out dramatically over our sample period, even when compared to the 2008 realization and all other troughs in the time series. Furthermore, several downward liquidity shocks have also been registered in the decade which has led to the 2008 crisis. Collected together, our results which focus on ex-post pricing conditional on the factor mean, suggest that it is the *extreme negative* liquidity shocks which hold identifying information on its pricing influence.

It is important to qualify the above discussion given the outcomes of tests on $\gamma_c^* = 0$: interestingly, we reject this hypothesis in 1981–90 and 1991–2000, and these are the same subperiods where evidence of factor pricing is detected. One may thus conclude that the market factor fails to satisfy the model pricing restrictions. Further analysis reveals that this is not the whole story. When the exercise is repeated imposing $\gamma_c^* = 0$ [refer to the supplementary Appendix], the confidence sets are empty in 1981–90 and 1991–2000. This suggests misspecification more broadly. Our results with missing factors reinforce such an interpretation, since all confidence sets in Panel B of Table 2 are completely uninformative.

Table 3 next assesses whether the momentum factor affects our analysis when added to the same set of factors. This table imposes $\gamma_c^* = 0$, since our analysis with an unrestricted cross-sectional intercept [refer to the supplementary Appendix] produces $\mathbb{R}$

Table 3
Liquidity, oil and momentum risk factors.

$r_i - \iota_T \gamma_0 = (\mathcal{R}_1 - \iota_T \gamma_0) b_{1i} + (F - \iota_T \gamma'_F) b_{iF} + u_i, \quad i = 1, \dots, n$								
$\theta = (\gamma_0, \gamma'_F) = (\theta_{\text{MKT}}, \theta_{\text{IAL}}, \theta_{\text{SCOP}}, \theta_{\text{MOM}})$								
PANEL A: inverting Hotelling F-test								
$\times 10^{-4}$	$\mathcal{R}_1$		F					
	$\overline{F}$	θ_{MKT}	$\overline{F}$	θ_{IAL}	$\overline{F}$	θ_{SCOP}	$\overline{F}$	θ_{MOM}
61–70	38	–	–	–	–	–	73	–
71–80	30	$\mathbb{R}$	–124	$\mathbb{R}$	195	$\mathbb{R}$	113	$\mathbb{R}$
81–90	44	$\mathbb{R}$	13	$]-\infty, -765]$ $\cup [-407, \infty[$	–25	$\mathbb{R}$	66	$]-\infty, 10]$ $\cup [353, \infty[$
91–00	103	$\mathbb{R}$	3	$]-\infty, -351]$ $\cup [986, \infty[$	3	$\mathbb{R}$	112	$]-\infty, -89]$ $\cup [303, \infty[$
01–10	14	$\mathbb{R}$	–26	$]-\infty, -351]$ $\cup [357, \infty[$	95	$\mathbb{R}$	–3	$\mathbb{R}$
All	46	$\mathbb{R}$	–41	$\mathbb{R}$	67	$\mathbb{R}$	69	$\mathbb{R}$
PANEL B: inverting mispricing robust F-test								
61–70	38	–	–	–	–	–	73	–
71–80	30	$\mathbb{R}$	–124	$\mathbb{R}$	195	$\mathbb{R}$	113	$\mathbb{R}$
81–90	44	$\mathbb{R}$	13	$\mathbb{R}$	–25	$\mathbb{R}$	66	$\mathbb{R}$
91–00	103	$\mathbb{R}$	3	$\mathbb{R}$	3	$\mathbb{R}$	112	$\mathbb{R}$
01–10	14	$\mathbb{R}$	–26	$\mathbb{R}$	95	$\mathbb{R}$	–3	$\mathbb{R}$
All	46	$\mathbb{R}$	–41	$\mathbb{R}$	67	$\mathbb{R}$	69	$\mathbb{R}$

Note — See notes to Table 2. Confidence sets are based on (3.16), in their Hotelling complete model form in Panel A (see Section 3), and their missing factor counterpart in Panel B (see Section 4).

Table 4
Size and book-to-market risk factors, unrestricted inference.

$r_i - \mathcal{R}_1 = \gamma_c^* \iota_T + (\mathcal{R}_1 - \gamma_0 \iota_T) d_i + (F - \iota_T \gamma'_F) b_{iF} + u_i, \quad i = 1, \dots, n$							
$\gamma_c^* = \gamma_c - \gamma_0, \quad \theta = (\gamma_0, \gamma'_F) = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}})$							
PANEL A: inverting Hotelling F-test							
$\times 10^{-4}$	CTE	$\mathcal{R}_1$		F			
	γ_c^*	$\overline{F}$	θ_{MKT}	$\overline{F}$	θ_{SMB}	$\overline{F}$	θ_{HML}
61–70	[–8, 101]	38	[–777, 9]	33	[–17, 94]	53	[–81, 82]
71–80	[5, 217]	30	[–715, –35]	43	[–34, 178]	33	[–179, 173]
81–90	$\mathbb{R}$	44	$]-\infty, -164]$ $\cup [707, \infty[$	–16	$\mathbb{R}$	56	$\mathbb{R}$
91–00	$]-\infty, -898]$ $\cup [150, \infty[$	103	$]-\infty, -365]$ $\cup [1950, \infty[$	4	$]-\infty, -964]$ $\cup [58, \infty[$	29	$\mathbb{R}$
00–10	[–1876, 115]	14	[–121, 2768]	57	[–2461, 84]	40	[–254, 245]
All	[–15, 120]	46	[–180, 139]	24	[–37, 103]	42	[16, 130]
PANEL B: inverting mispricing robust F-test							
61–70	[–16, 179]	38	[–2497, 50]	33	[–19, 196]	53	[–364, 56]
71–80	$\mathbb{R}$	30	$]-\infty, -54]$ $\cup [1211, \infty[$	43	$\mathbb{R}$	33	$]-\infty, 228]$ $\cup [1068, \infty[$
81–90	$\mathbb{R}$	44	$]-\infty, 153]$ $\cup [505, \infty[$	–16	$\mathbb{R}$	56	$\mathbb{R}$
91–00	$\mathbb{R}$	103	$\mathbb{R}$	4	$\mathbb{R}$	29	$\mathbb{R}$
00–10	$\mathbb{R}$	14	$\mathbb{R}$	57	$\mathbb{R}$	40	$\mathbb{R}$
All	[–13, 171]	46	[–305, 156]	24	[–35, 164]	42	[16, 141]

Note — See notes to Table 2.

almost everywhere. In contrast, imposing $\gamma_c^* = 0$ while interpreting the model as a complete specification reveals noteworthy results. Momentum is priced in 1981–90 and 1991–2000, and these are the same subperiods where the market was priced in our analysis excluding momentum; yet the market factor is not identified in the presence of momentum, and the same holds for oil price, since we obtain the real sets as an outcome on both pricing parameters over all subperiods and the full sample. In contrast, liquidity is priced in 1991–2000 and 2000–2010. As with our exercise without momentum, we obtain the real set everywhere when the model is viewed as an incomplete specification.

Taken collectively, our results make several contributions to the literature on liquidity and oil price. *First*, Pástor and Stambaugh (2019) underscore the superiority of their proposed non-traded factor, relative to its traded counterpart. However, we find that important identification costs are associated with their preferred measure. *Second*, Pástor and Stambaugh (2003) provide asset pricing theory and empirical links between liquidity and momentum; a more recent literature reviewed by Pástor and Stambaugh

Table 5

Size and book-to-market risk factors, zero-beta rate partialled out.

$r_i - r_n = (\mathcal{R}_1 - \gamma_0 \iota_T) b_{i1}^* + (F - \iota_T \gamma_F') b_{iF}^* + u_i^*$						
$\theta = (\gamma_0, \gamma_F') = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}})$, γ_c partialled-out						
PANEL A: inverting partial Hotelling F-test						
$\times 10^{-4}$	$\mathcal{R}_1$ $\overline{F}$	θ_{MKT}	F $\overline{F}$	θ_{SMB}	$\overline{F}$	θ_{HML}
61–70	38	[−642, −14]	33	[−11, 83]	53	[−64, 74]
71–80	30	[−621, −54]	43	[−26, 158]	33	[−137, 160]
81–90	44	$]-\infty, -198]$ $\cup [807, \infty[$	−16	$\mathbb{R}$	56	$\mathbb{R}$
91–00	103	$]-\infty, -432]$ $\cup [3087, \infty[$	4	$]-\infty, -1499]$ $\cup [97, \infty[$	29	$]-\infty, -281]$ $\cup [-24, \infty[$
00–10	14	[−100, 1226]	57	[−1089, 66]	40	[−137, 192]
All	46	[−170, 131]	24	[−34, 98]	42	[18, 127]
PANEL B: inverting partial mispricing robust F-test						
61–70	38	$]-\infty, 182]$ $\cup [1202, \infty[$	33	$]-\infty, -61]$ $\cup [-35, \infty[$	53	$\mathbb{R}$
71–80	30	$]-\infty, 55]$ $\cup [126, \infty[$	43	$\mathbb{R}$	33	$]-\infty, 361]$ $\cup [1184, \infty[$
81–90	44	$]-\infty, -108]$ $\cup [726, \infty[$	−16	$\mathbb{R}$	56	$\mathbb{R}$
91–00	103	[276, 2327]	4	[−751, 9]	29	[10, 228]
00–10	14	[−206, 197]	57	[−163, 155]	40	[−100, 98]
All	46	[−210, 143]	24	[−20, 101]	42	[18, 122]

Note — See notes to Table 2. Confidence sets invert the test based on (3.24), in its Hotelling complete model form in Panel A (see Section 3), and its missing factor counterpart in Panel B (see Section 4).

Table 6

Size, book to market and momentum risk factors.

$r_i - \iota_T \gamma_0 = (\mathcal{R}_1 - \iota_T \gamma_0) b_{i1} + (F - \iota_T \gamma_F') b_{iF} + u_i, \quad i = 1, \dots, n$								
$\theta = (\gamma_0, \gamma_F') = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}, \theta_{\text{MOM}})$								
PANEL A: inverting Hotelling F-test								
$\times 10^{-4}$	$\mathcal{R}_1$ $\overline{F}$	θ_{MKT}	F $\overline{F}$	θ_{SMB}	$\overline{F}$	θ_{HML}	$\overline{F}$	θ_{MOM}
61–70	38	$\mathbb{R}$	33	$\mathbb{R}$	53	$\mathbb{R}$	73	$\mathbb{R}$
71–80	30	[−300, 232]	43	[−62, −17]	33	[−87, 111]	113	[−333, 98]
81–90	44	$\mathbb{R}$	−16	$\mathbb{R}$	56	$\mathbb{R}$	66	$\mathbb{R}$
91–00	103	[−617, 123]	4	[−387, −20]	29	[−267, 36]	112	[−995, −45]
00–10	14	[−124, 258]	57	[−212, 5]	40	[−52, 204]	−3	[−554, 129]
All	46	[−3, 142]	24	[−46, 16]	42	[17, 80]	72	[−75, 219]
PANEL B: inverting mispricing robust F-test								
61–70	38	$\mathbb{R}$	33	$\mathbb{R}$	53	$\mathbb{R}$	73	$\mathbb{R}$
71–80	30	$\mathbb{R}$	43	$\mathbb{R}$	33	$\mathbb{R}$	113	$\mathbb{R}$
81–90	44	$\mathbb{R}$	−16	$\mathbb{R}$	56	$\mathbb{R}$	66	$\mathbb{R}$
91–00	103	$\mathbb{R}$	4	$\mathbb{R}$	29	$\mathbb{R}$	112	$\mathbb{R}$
00–10	14	[−334, 579]	57	[−707, 19]	40	[−106, 450]	−3	[−1837, 127]
All	46	[−45, 166]	24	[−80, 8]	42	[10, 81]	72	[−236, 103]

Note — See notes to Table 3.

(2019) supports these linkages. Along the same lines, Pástor and Stambaugh (2019) highlight the relevance of the liquidity factor in question during financial crises or economic disruptions. Similar arguments prevail on oil shocks; see Chiang et al. (2015) for a recent perspective with reference to asset pricing. We find that such linkages can compromise identification of the factors' individual pricing parameters. Third, Chiang et al. (2015) motivate their nonlinear factor model as a response to the state of the literature that has not led to a consensus view on the role of oil in pricing non-oil assets. Our identification-robust analysis provides a new take on this problem: what seems to drive identification are extreme disruptions. Since severe shocks (naturally) erupt in some though not all time periods, it will be much more arduous to identify pricing influence in decades where, as expressed by Pástor and Stambaugh (2019) “the factor sleeps”.

Let us now turn to the Fama–French model. We first consider the case where $\gamma_c^* \neq 0$, and θ_{SMB} and θ_{HML} are free parameters; refer to Table 4. Over the full sample period, the confidence sets with and without missing factors are not empty yet are quite wide; tests on the factor price and intercept are insignificant. Turning to subperiod analysis, when the model is treated as a complete

Table 7
Size, book to market and liquidity risk factors.

$r_i - \iota_T \gamma_0 = (\mathcal{R}_1 - \iota_T \gamma_0) b_{1i} + (F - \iota_T \gamma'_F) b_{iF} + u_i, \quad i = 1, \dots, n$								
$\theta = (\gamma_0, \gamma'_F) = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}, \theta_{\text{IAL}})$								
PANEL A: inverting Hotelling F-test								
$\times 10^{-4}$	$\mathcal{R}_1$		F		F		F	
	$\bar{F}$	θ_{MKT}	$\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}	$\bar{F}$	θ_{IAL}
61–70	38	–	33	–	53	–	–	–
71–80	30	$\mathbb{R}$	43	$\mathbb{R}$	33	$\mathbb{R}$	–124	$]-\infty, -458]$ $\cup [-143, \infty[$
81–90	44	$]-\infty, 32]$ $\cup [957, \infty[$	–16	$\mathbb{R}$	56	$]-\infty, 43]$ $\cup [530, \infty[$	13	$\mathbb{R}$
91–00	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$	3	$\emptyset$
00–10	14	$\mathbb{R}$	57	$\mathbb{R}$	40	$\mathbb{R}$	–26	$\mathbb{R}$
All	46	$\mathbb{R}$	19	$\mathbb{R}$	43	$\mathbb{R}$	–41	$\mathbb{R}$
PANEL B: inverting mispricing robust F-test								
61–70	38	$\mathbb{R}$	33	$\mathbb{R}$	53	$\mathbb{R}$	–3	$\mathbb{R}$
71–80	30	$\mathbb{R}$	43	$\mathbb{R}$	33	$\mathbb{R}$	–124	$\mathbb{R}$
81–90	44	$\mathbb{R}$	–16	$\mathbb{R}$	56	$\mathbb{R}$	13	$\mathbb{R}$
91–00	103	$\mathbb{R}$	4	$\mathbb{R}$	29	$\mathbb{R}$	3	$\mathbb{R}$
00–10	14	$\mathbb{R}$	57	$]-\infty, 30]$ $\cup [120, \infty[$	40	$]-\infty, -317]$ $\cup [-113, \infty[$	–26	$]-\infty, 348]$ $\cup [2183, \infty[$
All	46	$\mathbb{R}$	19	$\mathbb{R}$	43	$\mathbb{R}$	–41	$\mathbb{R}$

Note — See notes to Table 3.

specification, the market factor is priced until 2000, despite the prevalence of unbounded outcomes; see Harvey and Liu (2021) for a recent perspective on the market factor. Size is the only other factor which is priced, and this occurs only in the 1991–2000 subperiod. However, we find that the cross-sectional intercept differs from zero in two subperiods, namely 1971–80 and 1991–2000. When inference is repeated imposing a zero cross-sectional intercept [results not reported for space considerations], the sets are empty in both subperiods. The cross-sectional intercept thus reveals crucial information on model fit, the interpretation of which goes beyond implications for traded factors. We thus ask whether viewing this model as a complete specification with a missing factor improves pricing outcomes. Panel B confirms that this is not the case, given the severely uninformative sets we obtain.

This motivates an evaluation of whether the partialled-out test which evacuates the intercept as in Kleibergen and Zhan (2020) holds more information on this model. We thus report the results of this method in Panel A of Table 5; Panel B of this table reports our extension of the method that allows for missing factors. Despite partialling-out, our main findings on pricing are not altered relative to Table 4, except that the pricing influence of the market factor seems weaker, whether we consider the model as complete or not. Basically, we do not find that identification improves through the partialled-out approach, which confirms our simulation results. Recall that crucially, this approach forgoes the information conveyed by the cross-sectional intercept on model fit. We find similar evidence with partialled-out procedures for all the models we analyse. These are not reported for space considerations; instead, we report Table 5 for illustrative purposes.

We next expand the Fama–French specification in two directions: (i) we add momentum, which is a common practice (Fama and French, 2018); we also consider the addition of a non-tradable factor, to qualify our findings with momentum; (ii) we assess and impose tradability restrictions on all factors, and then build on the fact that size, book-to-market and momentum are return spreads (Shanken and Weinstein, 2006), so their price is fixed prior to estimation and testing, as equal to the factor mean.

Results which include momentum are reported in Table 6, where $\gamma_c^* = 0$, and θ_{SMB} , θ_{HML} and θ_{MOM} are free parameters. This table imposes $\gamma_c^* = 0$ since our results with an unrestricted cross-sectional intercept reveal non-informative sets everywhere. We see from this table that imposing $\gamma_c^* = 0$ upholds the model as a complete specification, in contrast to the case excluding momentum where we obtain empty sets in the 1970s and 1990s. However, the prevalence of uninformative sets across subperiods casts doubt on the conclusions one can draw from the fact that as a whole, the model with momentum survives rejection. Said differently, we hesitate to conclude that the model excluding momentum is rejected in favour of the augmented counterpart, since identification remains weak with the latter. *The most noteworthy result is that momentum is priced in the 1970s and 1990s, and these are the decades where the three-factor specification is rejected.* The pricing influence of the market is no longer significant, whereas size is priced in several subperiods as well as over the full sample. From Panel B of this table, we find that allowing for a missing factor yields $\mathbb{R}$ everywhere, except in the last decade and over the full sample in which case size is the only priced factor.

Table 7 reports our results with the addition of liquidity [refer to the supplementary Appendix for results on the addition of oil, which add no further insights to our discussion]. In contrast to momentum, the model treated as a complete specification is rejected in the 90s, whereas liquidity restores the significance of the market factor in the 80s. Yet as we will show next, results with liquidity are critically different when further model-consistent restrictions are imposed.

So following the road map set above, we now reconsider the three-factor Fama–French model where $\gamma_c^* \neq 0$, and $\theta_{\text{SMB}} = \theta_{\text{HML}} = 0$, which stipulates that SMB and HML are tradable return spreads; see Table 8. These restrictions are rejected over the full sample period, and in the 1980s and 90s. Furthermore, the confidence set for the cross-sectional intercept in the 70s exclude zero; when

Table 8

Size and book-to-market risk factors, restricted return spreads.

$r_i - \bar{R}_i = \gamma_c^* t_T + (\bar{R}_i - \gamma_0 t_T) d_i + (F - t_T \gamma_F') b_{iF} + u_i, \quad i = 1, \dots, n$							
$\gamma_c^* = \gamma_c - \gamma_0, \theta = (\gamma_0, \gamma_F') = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}), \theta_{\text{HML}} = \theta_{\text{SMB}} = 0$							
PANEL A: inverting Hotelling F-test							
$\times 10^{-4}$	CTE γ_c^*	$\bar{R}_i$ $\bar{F}$	θ_{MKT}	F $\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}
61–70	[−6, 31]	38	[−595, −78]	33	0	53	0
71–80	[21, 67]	30	[−453, −129]	43	0	33	0
81–90	$\emptyset$	44	$\emptyset$	−16	0	56	0
91–00	$\emptyset$	103	$\emptyset$	4	0	29	0
01–10	[−1, 56]	14	[−107, 49]	57	0	40	0
All	$\emptyset$	46	$\emptyset$	24	0	42	0
PANEL B: inverting mispricing robust F-test							
61–70	[−1, 29]	38	[−585, 30]	33	0	53	0
71–80	[36, 61]	30	[−566, −358]	43	0	33	0
81–90	$\emptyset$	44	$\emptyset$	−16	0	56	0
91–00	[−6, 21]	103	[−361, −115]	4	0	29	0
01–10	[−7, 68]	14	[−159, 146]	57	0	40	0
All	$\emptyset$	46	$\emptyset$	24	0	42	0

Note — See notes to Table 2. The inverted test stipulates that SMB and HML are tradable return spreads.

Table 9

Size and book-to-market risk factors, restricted as tradable.

$r_i - \bar{R}_i = \gamma_c^* t_T + (\bar{R}_i - \gamma_0 t_T) b_i^* + (F - t_T \gamma_F') b_{iF}^* + u_i, \quad i = 1, \dots, n$							
$\gamma_c^* = \gamma_c - \gamma_0, \theta = (\gamma_0, \gamma_F') = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}), \theta_{\text{MKT}} = \theta_{\text{SMB}} = \theta_{\text{HML}} = \gamma_0$							
PANEL A: inverting Hotelling F-test							
$\times 10^{-4}$	CTE γ_c^*	$\bar{R}_i$ $\bar{F}$	θ_{MKT}	F $\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}
61–70	$\emptyset$	38	$\emptyset$	33	$\emptyset$	53	$\emptyset$
71–80	$\emptyset$	30	$\emptyset$	43	$\emptyset$	33	$\emptyset$
81–90	$\emptyset$	44	$\emptyset$	−16	$\emptyset$	56	$\emptyset$
91–00	$\emptyset$	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$
01–10	[−14, 50]	14	[−43, 35]	57	[−43, 35]	40	[−43, 35]
All	$\emptyset$	46	$\emptyset$	24	$\emptyset$	42	$\emptyset$
PANEL B: inverting mispricing robust F-test							
61–70	[1, 31]	38	[−7, 21]	33	[−7, 21]	53	[−7, 21]
71–80	$\emptyset$	30	$\emptyset$	43	$\emptyset$	33	$\emptyset$
81–90	$\emptyset$	44	$\emptyset$	−16	$\emptyset$	56	$\emptyset$
91–00	$\emptyset$	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$
01–10	[−85, 130]	14	[−106, 89]	57	[−106, 89]	40	[−106, 89]
All	$\emptyset$	46	$\emptyset$	24	$\emptyset$	42	$\emptyset$

Note — See notes to Table 2. The inverted test stipulates that SMB and HML are tradable.

inference is repeated imposing a zero cross-sectional intercept [results not reported for space considerations], the model is rejected in this decade as well. Again, this reinforces the role of our test on the model intercept. The last decade still yields uninformative results. *The most notable result emerges from Panel B: when one allows for a missing factor, the restriction associated with tradable and return spread factors visibly improves identification (with reference to all previous tables).* In this case, the market is priced except in the last decade that still yields uninformative outcomes, and rejections of the model are upheld only in the 80s and over the full sample. Table 9 takes a different model-consistent restriction to the data, namely that $\gamma_c^* \neq 0$, and $\theta_{\text{SMB}} = \theta_{\text{HML}} = \theta_{\text{MKT}} = \gamma_0$, which stipulates that SMB and HML are tradable ignoring the return spread property. These restrictions leads to model rejection over the full sample period, and in all subperiods except the last decade; in this case, the size and book to market factors are both priced. *Here again, we find that the model interpreted with a missing factor is better identified than when the restriction is ignored.* All three factors are priced in the first decade, although the first and last decades yield uninformative outcomes and rejections are upheld over the remaining decades and over the full sample.

Importantly, these results illustrate the identification versus misspecification tensions which stem from the considered restrictions. This reflects a key model building reality; should the LAPM be considered in its unrestricted form with free parameters, or should model-consistent restrictions be accounted for? Our contribution is to provide simultaneous inference on pricing and the validity of these restrictions (through the cross-sectional intercept) that yields empirical guidance towards managing these tensions.

Table 10

Size, book-to-market and momentum risk factors, restricted as spreads.

$$r_i - \bar{R}_1 = \gamma_c^* t_T + (\bar{R}_1 - \gamma_0 t_T) d_i + (F - t_T \gamma_F') b_{iF} + u_i, \quad i = 1, \dots, n$$

$$\gamma_c^* = \gamma_c - \gamma_0, \quad \theta = (\gamma_0, \gamma_F') = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}, \theta_{\text{MOM}}), \quad \theta_{\text{HML}} = \theta_{\text{MKT}} = \theta_{\text{MOM}} = 0$$

PANEL A: inverting Hotelling F-test									
$\times 10^{-4}$	CTE	$\bar{R}_1$	θ_{MKT}	$\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}	$\bar{F}$	θ_{MOM}
	γ_c^*	$\bar{F}$							
61–70	[−5, 53]	38	[−789, −3]	33	0	53	0	73	0
71–80	[16, 72]	30	[−408, −37]	43	0	33	0	113	0
81–90	$\emptyset$	44	$\emptyset$	−16	0	56	0	66	0
91–00	[52, 140]	103	[−527, −172]	4	0	29	0	112	0
01–10	[−2, 64]	14	[−137, 53]	57	0	40	0	−3	0
All	$\emptyset$	46	$\emptyset$	24	0	42	0	72	0
PANEL B: inverting mispricing robust F-test									
$\times 10^{-4}$	γ_c^*	$\bar{F}$	θ_{MKT}	$\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}	$\bar{F}$	θ_{MOM}
61–70	[1, 43]	38	[−645, 121]	33	0	53	0	73	0
71–80	[23, 57]	30	[−397, −109]	43	0	33	0	113	0
81–90	[7, 38]	44	[−520, −240]	−16	0	56	0	66	0
91–00	[2, 105]	103	[−818, 62]	4	0	29	0	112	0
01–10	[1, 71]	14	[−267, 119]	57	0	40	0	−3	0
All	$\emptyset$	46	$\emptyset$	24	0	42	0	72	0

Note — See notes to tables [Table 8](#).**Table 11**

Size, book-to-market and momentum risk factors, restricted as tradable.

$$r_i - \bar{R}_1 = \gamma_c^* t_T + (\bar{R}_1 - \gamma_0 t_T) d_i + (F - t_T \gamma_F') b_{iF} + u_i, \quad i = 1, \dots, n$$

$$\gamma_c^* = \gamma_c - \gamma_0, \quad \theta = (\gamma_0, \gamma_F') = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}, \theta_{\text{MOM}}), \quad \theta_{\text{MKT}} = \theta_{\text{SMB}} = \theta_{\text{HML}} = \theta_{\text{MOM}} = \gamma_0$$

PANEL A: inverting Hotelling F-test									
$\times 10^{-4}$	CTE	$\bar{R}_1$	θ_{MKT}	$\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}	$\bar{F}$	θ_{MOM}
	γ_c^*	$\bar{F}$							
61–70	[2, 32]	38	[1, 31]	33	[1, 31]	53	[1, 31]	73	[1, 31]
71–80	$\emptyset$	30	$\emptyset$	43	$\emptyset$	33	$\emptyset$	113	$\emptyset$
81–90	$\emptyset$	44	$\emptyset$	−16	$\emptyset$	56	$\emptyset$	66	$\emptyset$
91–00	$\emptyset$	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$	112	$\emptyset$
01–10	[−18, 48]	14	[−53, 35]	57	[−53, 35]	40	[−53, 35]	−3	[−53, 35]
All	$\emptyset$	46	$\emptyset$	24	$\emptyset$	42	$\emptyset$	72	$\emptyset$
PANEL B: inverting mispricing robust F-test									
$\times 10^{-4}$	γ_c^*	$\bar{F}$	θ_{MKT}	$\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}	$\bar{F}$	θ_{MOM}
61–70	[−10, 45]	38	[−24, 30]	33	[−24, 30]	53	[−24, 30]	73	[−24, 30]
71–80	$\emptyset$	30	$\emptyset$	43	$\emptyset$	33	$\emptyset$	113	$\emptyset$
81–90	[−143, −30]	44	[−155, −28]	−16	[−155, −28]	56	[−155, −28]	66	[−155, −28]
91–00	$]-\infty, 55]$ $\cup [467, \infty[$	103	$]-\infty, 30]$ $\cup [611, \infty[$	4	$]-\infty, 30]$ $\cup [611, \infty[$	29	$]-\infty, 30]$ $\cup [611, \infty[$	112	$]-\infty, 30]$ $\cup [611, \infty[$
01–10	[−73, 81]	14	[−130, 56]	57	[−130, 56]	40	[−130, 56]	−3	[−130, 56]
All	$\emptyset$	46	$\emptyset$	24	$\emptyset$	42	$\emptyset$	72	$\emptyset$

Note — See notes to tables [Table 9](#).

To complete our analysis, we introduce (in turn) momentum, liquidity and oil, imposing each of the above restrictions. In [Table 10](#), $\gamma_c^* \neq 0$, and $\theta_{\text{SMB}} = \theta_{\text{HML}} = \theta_{\text{MOM}} = 0$, which stipulates that SMB, HML and MOM are tradable return spreads. Despite lack of fit over the full sample, the model with or without a missing factor appears to fare better than the three-factor one.

Turning to [Table 11](#) where $\gamma_c^* \neq 0$, and $\theta_{\text{MKT}} = \theta_{\text{SMB}} = \theta_{\text{HML}} = \theta_{\text{MOM}} = \gamma_0$ (which entails that all factors are tradable), we again observe that the model is much better identified than when the restriction is ignored, particularly when interpreted with a missing factor. Importantly, the addition of momentum does not entail identification cost, at least relative to the unrestricted specification. With or without a missing factor, the model fares well in the 60s, and all factors including momentum are priced. When a missing factor is allowed, the model also fares well in the 80s and 90s, and both market and momentum are priced, yet γ_c^* appears to be significantly different from zero in the 80s; when inference is repeated imposing $\gamma_c^* = 0$ [refer to the supplementary Appendix], the model is rejected in the 80s. Overall, and relative to all previous specifications, this tradable specification including momentum is more informative on factor pricing, particularly when we allow for a missing factor. We still observe rejections over the full sample period and uninformative outcomes in

Table 12

Size, book-to-market and oil risk; MKT, SMB, HML restricted as tradable.

$$r_i - \iota_T \gamma_0 = (\mathcal{R}_1 - \gamma_0 \iota_T) b_{i1} + (F - \iota_T \gamma'_F) b_{iF} + u_i, \quad i = 1, \dots, n$$

$$\theta = (\gamma_0, \gamma'_F) = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}, \theta_{\text{SCOP}}), \quad \theta_{\text{MKT}} = \theta_{\text{SMB}} = \theta_{\text{HML}} = \gamma_0$$

PANEL A: inverting Hotelling F-test								
$\times 10^{-4}$	$\mathcal{R}_1$		F		$\bar{F}$		$\bar{F}$	
	$\bar{F}$	θ_{MKT}	$\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}	$\bar{F}$	θ_{SCOP}
61–70	38	–	33	–	53	–	–	–
71–80	30	[–12, 75]	43	[–12, 75]	33	[–12, 75]	195	[–9497, –1485]
81–90	44	$\emptyset$	–16	$\emptyset$	56	$\emptyset$	–25	$\emptyset$
91–00	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$	3	$\emptyset$
01–10	14	[–127, 0]	57	[–127, 0]	40	[–127, 0]	95	[–106, 1569]
All	46	$\emptyset$	19	$\emptyset$	43	$\emptyset$	67	$\emptyset$
PANEL B: inverting mispricing robust F-test								
61–70	38	–	33	–	53	–	–	–
71–80	30	$\mathbb{R}$	43	$\mathbb{R}$	33	$\mathbb{R}$	195	$] -\infty, -1712]$ $\cup [3122, \infty[$
81–90	44	[–18, 29]	–16	[–18, 29]	56	[–18, 29]	–25	[196, 658]
91–00	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$	3	$\emptyset$
01–10	14	[–62, 51]	57	[–62, 51]	40	[–62, 51]	95	[–831, 389]
All	46	$\emptyset$	19	$\emptyset$	43	$\emptyset$	67	$\emptyset$

Note — See notes to Table 9.

Table 13

Size, book-to-market and liquidity risk; MKT, SMB, HML restricted as tradable.

$$r_i - \iota_T \gamma_0 = (\mathcal{R}_1 - \gamma_0 \iota_T) b_{i1} + (F - \iota_T \gamma'_F) b_{iF} + u_i, \quad i = 1, \dots, n$$

$$\theta = (\gamma_0, \gamma'_F) = (\theta_{\text{MKT}}, \theta_{\text{SMB}}, \theta_{\text{HML}}, \theta_{\text{IAL}}), \quad \theta_{\text{MKT}} = \theta_{\text{SMB}} = \theta_{\text{HML}} = \gamma_0$$

PANEL A: inverting Hotelling F-test								
$\times 10^{-4}$	$\mathcal{R}_1$		F		$\bar{F}$		$\bar{F}$	
	$\bar{F}$	θ_{MKT}	$\bar{F}$	θ_{SMB}	$\bar{F}$	θ_{HML}	$\bar{F}$	θ_{IAL}
61–70	38	–	33	–	53	–	–	–
71–80	30	$\mathbb{R}$	43	$\mathbb{R}$	33	$\mathbb{R}$	–124	$] -\infty, -1024]$ $\cup [214, \infty[$
81–90	44	[–14, 10]	–16	[–14, 10]	56	[–14, 10]	13	[–609, –318]
91–00	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$	3	$\emptyset$
01–10	14	$\mathbb{R}$	57	$\mathbb{R}$	40	$\mathbb{R}$	–26	$] -\infty, -576]$ $\cup [-430, \infty[$
All	46	$\mathbb{R}$	24	$\mathbb{R}$	42	$\mathbb{R}$	–41	$] -\infty, -2733]$ $\cup [340, \infty[$
PANEL B: inverting mispricing robust F-test								
61–70	38	–	33	–	53	–	–	–
71–80	30	$\mathbb{R}$	43	$\mathbb{R}$	33	$\mathbb{R}$	–124	$] -\infty, -2992]$ $\cup [378, \infty[$
81–90	44	$\mathbb{R}$	–16	$\mathbb{R}$	56	$\mathbb{R}$	13	$] -\infty, -881]$ $\cup [628, \infty[$
91–00	103	$\emptyset$	4	$\emptyset$	29	$\emptyset$	3	$\emptyset$
01–10	14	[–78, 16]	57	[–78, 16]	40	[–78, 16]	–26	[–2495, 306]
All	46	$] -\infty, 6]$ $\cup [86, \infty[$	24	$] -\infty, 6]$ $\cup [86, \infty[$	42	$] -\infty, 6]$ $\cup [86, \infty[$	–41	$] -\infty, -5057]$ $\cup [186, \infty[$

Note — See notes to Table 9.

the last decade. The identification versus misspecification tensions thus remain albeit to a lesser extent, which in view of our results broadly is the most pervasive takeaway.

Results that study the addition of the oil factor, all other factors imposed as tradable are reported in Table 12; for more complete results refer to the supplementary Appendix. Despite model misspecification evidence, the key results which emerge from this table are in Panel B that allows for a missing factor; in this case, the oil factor is priced in the 70s and 80s, which viewed in conjunction with our results with momentum in Table 11, suggests that *an additional source of risk improves the specification of the Fama–French model*. This result is further confirmed in Table 13 which illustrates the addition of liquidity, here again with all other factors imposed as tradable. While the specification is still rejected in the 90s, we find that all four factors are priced in the 80s, a result

that occurred on rare occasions in all our analyses.¹³ Furthermore, over the whole sample period, when we allow for a missing factor, all four factors are priced, which is the only case where any model fared this well from a full-sample perspective.

Our results can be summarized as follows.

(1) In LAPMs, the extent restrictions on the zero-beta rate are integrated into the analysis has important implications on the identifying rank condition of LAPMs. Now since the available finite-sample motivated tests (Kleibergen et al., 2021) do not allow one to impose such restrictions, our proposed tests provide an interesting insight which is useful to asset pricing modelling, identification and testing. On the one hand, these restrictions yield dimensionality reduction; on the other hand, they help channel the identifying information on the zero-beta rate that can be extracted from the data with finite-sample reliability.

(2) Model fit, identification and pricing influence of several factors vary consequentially across sub-periods. When models are not rejected, factor prices are often weakly identified, but there is evidence of relatively short time spells where some, although not all, factors are priced. In addition, episodes where all factors in a model are priced are scarce. The short duration of these effects makes inference on the risk-return relationship particularly challenging. Temporal inconsistency is well known in asset pricing; leading arguments can be traced back to Huberman and Kandel (1987) (page 881), which support breaking analysis over “fairly short time periods”. We offer new insights to justify instabilities, that stem from the nature of the factors. Illustrative examples include the liquidity effects, the influence of which is only identified in decades which register disruptive events. We find that oil price as well as momentum display similar time varying exogenous variability, which translates into: (i) time varying identification strength for inference on pricing influence, and (ii) time varying evidence in favour of the restrictions which define the model. Searching for factors with significant pricing influence consistently may turn to be an elusive objective. Confidence sets which are empty in some subperiods and uninformative in others are also to be expected, within a LAPM motivated by (2.1).

(3) Bearing the above in mind, results indicate that a supplemental factor can improve the specification of the Fama–French model. Among the considered add-ons, namely momentum, oil and non-traded liquidity, the latter yields a better balance between identification and fit. The most important gains following the addition of liquidity show up over the entire time span, in which case all four factors turn out to be priced provided we account for tradable factor restrictions and, importantly, for a missing factor. Both momentum and liquidity are characteristics, which raises the enduring question on how to best capture such effects within the risk return relationship. For recent perspectives in this regard, see Kelly et al. (2019) or Kelly et al. (2021) who introduce a principal component approach using returns and a large data set of characteristics at the stock-level, to account for time varying exposure. In particular, Kelly et al. (2021) show, using various in and out-of-sample diagnostics, that such a methodology actually reveals a momentum premium, that is, an actual compensation for factor exposure. Although promising and economically relevant, such checks do not include an inferential assessment of associated risk price. Extensions in this direction are a promising research direction, possibly as in e.g. Giglio et al. (2023) who adapt principal components analysis for selection over test assets to address possible factor weakness.

(4) Finally, our study suggests that further research is called for to address the dimensionality problem associated with non-traded factors. Conditional models suffer from the same problem, as they typically lead to cross-product terms which increase dimensionality [see Beaulieu et al. (2007, 2016), Gospodinov et al. (2014) or the above cited framework of Kelly et al. (2021)]. Subset tests as in Kleibergen (2009) and Kleibergen and Zhan (2020) require a stance on the model to justify the estimation of non-tested parameters. Mimicking portfolios present useful opportunities yet raise error-in-variables challenges; see Kleibergen and Zhan (2018) for an identification-robust take on such an approach. Whether departing from traditional LAPM factor structures [as is e.g. Chiang et al. (2015) or Bandi et al. (2021)] provide another promising alternative is a worthy research avenue. Yet one must be mindful that identification concerns tend to increase with model complexity. We expect such general insights to be useful provided the profession aims for identification-robust analysis.

6. Conclusion

This paper proposes finite-sample identification-robust confidence sets for the zero-beta rate and factor price in linear asset pricing models. Our methodology is distinct along the following lines. Inference methods discriminate between tradable and non-tradable factors, and inform on factor price jointly with the cross-sectional intercept. Through the latter parameter, model-consistent restrictions which stem from the nature of the factors can be accounted for that materially affect outcomes. Mispricing is embedded within a complete linear asset pricing model albeit with an unobserved factor.

We apply the methodology proposed to specifications with: (i) Fama–French and momentum factors, and (ii) liquidity and oil price factors. Results underscore the identification versus misspecification trade-offs as restrictions that are compatible with asset pricing theory are factored in, and reinforce the usefulness of our missing factor approach in managing such trade-offs empirically. While issues with non-traded factors remain, this paper contributes to a promising research agenda towards robust factor discovery and opens up the way for model testing which accounts for missing factors.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jeconom.2024.105915>.

¹³ In addition to this case, all model factors are priced: (i) with the Fama–French model, in the 60s where only market is restricted as tradable, or in the 2000–10 decade when all factors are restricted as tradable, with and without a missing factor; (ii) with the Fama–French model augmented with momentum, in the 90s when all factors are restricted as tradable, including or excluding a missing source of risk though, with a restricted intercept; refer to the supplementary Appendix.

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